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# Fast and Stable Virtual Human Video Generation: A Survey

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**Abstract** Virtual-human video generation requires timely output and consistent identity, appearance, and motion. This survey examines these requirements through two perspectives, *Fast* and *Stable*, focusing on talking portraits and selected upper- and full-body systems. We classify methods into GAN, diffusion, visual-autoregressive, and rendering routes according to the component that produces appearance, while distinguishing control, state, and decoding in hybrid systems. We review acceleration through compact representations, fewer model evaluations, computation reuse, and execution optimization. We then discuss temporal failures and the roles of reference anchors, bounded memory, position handling, history-aware training, and boundary-state transfer. Representative studies connect these mechanisms to efficiency and output quality. We complement the review with single-H200 measurements of 56 configurations covering 55 methods, each measured three times after three complete same-interpreter warm-ups. Stage timings and warmed memory peaks reveal preparation, loading, decoding, and export costs beyond generation. Workloads and model-residency modes differ, so the results describe configurations rather than a controlled ranking or long-session stability. Finally, we review datasets, evaluation measures, and open problems in complete-pipeline measurement and sustained, controllable generation.

## 1 Introduction

Virtual-human video generation synthesizes a subject whose appearance and behavior follow speech, text, pose, expression, or video. A reference image, enrollment recording, or learned representation specifies the subject; the output may cover the face, upper body, or full body. Across these settings, the task is to preserve appearance while reproducing the requested motion [134, 183, 215, 240].

Deployment adds two requirements: computational efficiency within the available latency and resource budget [62, 81, 210], and consistency of identity, appearance, synchronization, and motion during sustained generation [113, 165]. We call these perspectives *Fast* and *Stable*. They require separate measurements and motivate our comparison of computation, temporal mechanisms, and deployment trade-offs.

Figure 1 distinguishes delayed output from unintended appearance changes; both must be evaluated while preserving requested motion.

Figure 2 groups representative mechanisms by route. Distillation reduces diffusion evaluations [127, 174, 227, 228]; caching reuses computation across steps or frames [118, 129, 132]; autoregression predicts visual state from cached history [27, 102, 209]; and rendering reuses enrolled appearance [62, 63, 206, 210]. These mechanisms reduce different costs. A local speedup may shift the bottleneck to conditioning, decoding, or serving, and throughput alone establishes neither first-output latency nor stability.

We organize this diversity by the component responsible for generated appearance, termed the *visual owner*. Identifying it gives a common basis for comparing computational costs and temporal behavior across tasks.

### 1.1 Scope: Fast and Stable

We review human-centered RGB generation with mechanisms for reducing runtime cost or supporting extended generation. Coverage centers on talking portraits and selected upper- and full-body systems, includ-

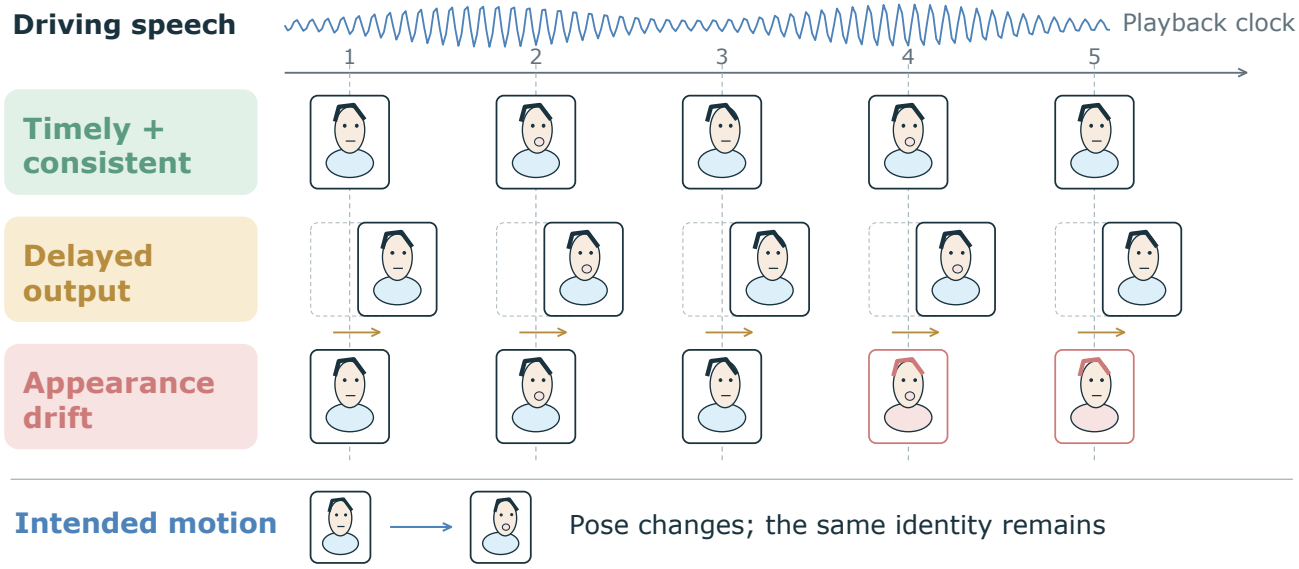



**Fig. 1** Fast and Stable requirements illustrated by aligned playback timelines. Delays affect responsiveness; unintended appearance changes affect consistency. Requested pose changes remain valid. Schematic portraits and timing marks.

ing generalizable and subject-specific methods. Neural radiance fields (NeRFs) and 3D Gaussian splatting (3DGS) are included when they support efficient appearance construction or frame production [91,137]. Motion-only, mesh, and coefficient predictors serve as supporting components of RGB pipelines.

Foundational references explain representations and mechanisms; contextual references clarify interfaces. Primary-route comparisons require a documented generation path and identifiable visual owner. Unsupported assignments are excluded, while missing timing or stability measurements are marked not reported (NR).

Two perspectives guide the analysis.

**Fast** concerns computation and time to visible output: representation size, model-call count, per-call cost, reuse, and decoding or rendering. Throughput and first-output latency retain the source's hardware and timing boundaries.

**Stable** concerns consistency over a continuous evaluated duration. We examine appearance drift and the memory, training, and recovery policies intended to control it. Demonstrated duration and quantitative stability evidence are recorded separately.

*Interactivity* is the ability to respond to new conditions, such as a dialogue turn, interruption, or pose change [81, 113, 218, 233, 237]. It does not imply long-session stability. We include interactive generators within our scope but leave interaction-specific rankings to their dedicated surveys and protocols [8, 15, 49, 231].

Efficiency and stability can favor different designs. Section 2 defines their common terms and measurement boundaries.

### 1.2 Literature Selection

We curate studies that clarify how appearance representation, computation reuse, or temporal state affects efficiency and stability. Coverage is selective rather than exhaustive, with architectural documentation required for primary-route assignment.

Foundational work supplies the technical background; motion- and coefficient-only models clarify control interfaces rather than constitute visual generators.

Releases and configurations are distinguished when architecture or results change. Source-reported measurements retain their settings, with NR for missing fields. Our measurements use separate tables and the protocol in Section 8.7.

### 1.3 Relation to Existing Surveys

Related surveys organize efficient generative modeling by distillation, caching, attention, and inference optimization [164]; long-video generation by temporal extension and resource or quality challenges [103]; and human generation by task or representation [134, 156, 161, 215]. Graphics reviews address avatar construction and appearance priors [266]. Interactive-video and avatar surveys emphasize the interaction loop and production requirements [49, 231].

Table 1 contrasts these emphases. Our review connects the appearance-producing component to its acceleration mechanisms and temporal behavior, making auxiliary roles in hybrid systems explicit.

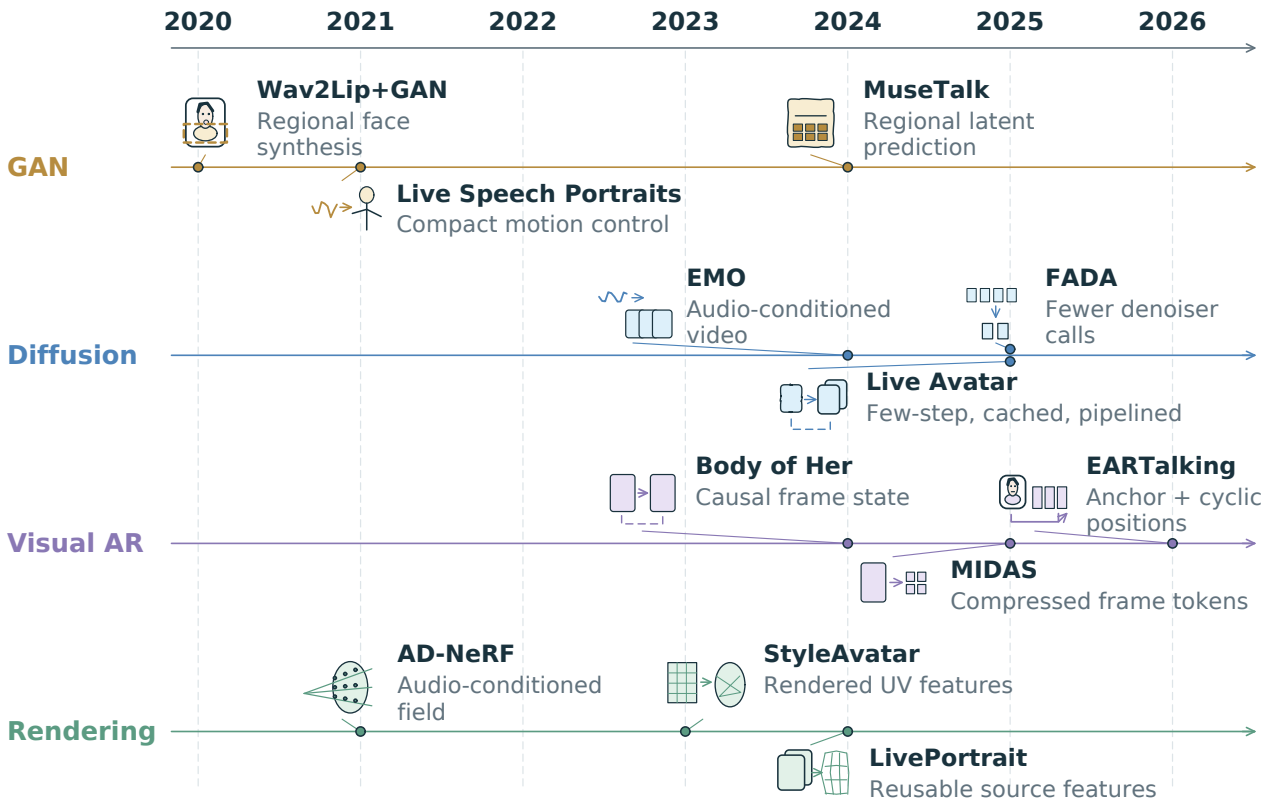



**Fig. 2** Selected mechanisms along the four routes: regional and latent face synthesis; fewer diffusion evaluations and pipelined serving; compressed causal visual state; and reusable fields or geometric appearance [5, 27, 62, 63, 81, 124, 150, 183, 194, 204, 242, 257]. Publication years identify the cited editions; the diagram does not rank performance.

Figure 3 relates this perspective to task- and efficiency-centered taxonomies.

### 1.4 Ownership Taxonomy

Figure 4 applies the ownership rules in order: rendering, visual autoregression, diffusion, then GAN synthesis. The precedence resolves hybrids, including autoregressive motion controllers coupled to subject-specific renderers. Section 2.3 defines the tests.

- **GAN-based methods** (Section 3) use an adversarially trained feed-forward generator for dense or regional imagery [58, 89, 150, 178, 242]. Regional prediction and compact control reduce per-frame work, while temporal consistency depends on the generator's conditioning and any temporal components.
- **Diffusion-based methods** (Section 4) use diffusion or flow matching to generate dense pixels, visual latents, or visual chunks [69, 73, 117, 183]. Acceleration reduces iterative calls, per-call cost, or repeated work; stability mechanisms include reference anchors, bounded memory, and training with generated history.
- **Autoregressive methods** (Section 5) predict evolving visual tokens or latents from previous visual state [5, 27, 204, 209]. Token volume and serial prediction determine much of the cost, while retained history can support continuity and propagate errors.
- **Render-based methods** (Section 6) reuse a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp [62, 91, 109, 137, 210]. Reused appearance can reduce repeated synthesis and anchor identity, with enrollment, tracking, deformation, and compositing contributing additional costs and failure modes.

Driving modality, anatomy, personalization, representation, and deployment mode are comparison axes. Auxiliary modules are annotated without adding another primary-route count.

### 1.5 Contributions and Main Findings

Our contributions are a taxonomy that resolves hybrid architectures, a cross-route analysis of acceleration and

**Table 1** Complementary survey perspectives, emphasizing the tasks, organizing principles, and practical questions addressed by each review.

| Survey / scope | Organizing principle | Efficiency and temporal emphasis | Relation to the present analysis |
|---|---|---|---|
| Deep Person Generation [161] | Face, pose, and clothing generation; shared technical components | Identity/attribute control, generation quality, and task-specific challenges | Connects human-generation tasks; our comparison unit is the deployed RGB pipeline. |
| Talking Head Synthesis [134] | Portrait generation, driving mechanisms, and editing | Task-dependent synthesis, control, and evaluation | Complements our separation of appearance, control, and persistent state. |
| Human Motion Video Generation [215] | Input, motion planning, video generation, refinement, and output; modality and task | Motion/video methods, datasets, evaluation, and temporal challenges | Provides broad human-motion coverage; we focus on cost, state, enrollment, and runtime and temporal behavior. |
| Efficient Diffusion Models [164] | Algorithm, system, and framework levels | Sampling and implementation efficiency across diffusion applications | Supplies acceleration concepts; we track which stage is accelerated and include GAN, visual AR, and rendering pipelines. |
| Interactive Real-Time Visual Avatar Generation [49] | Method families and latency, anatomy, personalization, and input modality | Production constraints and head–body integration; drift as a challenge | Closely related deployment perspective; our analysis connects module roles with computation and temporal behavior. |
| This survey | Visual owner plus independent control/state/decoder roles | Fast/Stable lenses; timing boundaries, enrollment, seven stability dimensions | Cross-route mechanism analysis; repeated runtime/memory measurements of 56 configurations; separate temporal evaluation. |

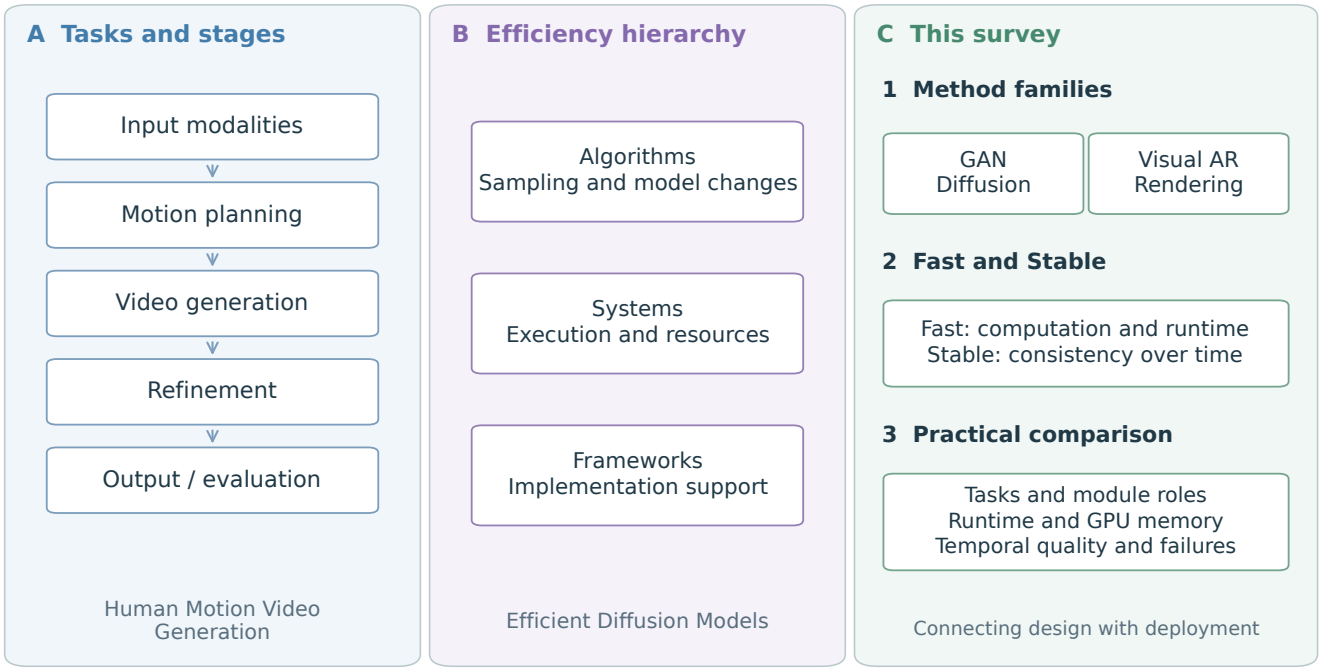



**Fig. 3** Complementary views of the literature: human-motion tasks and generation stages [215]; diffusion efficiency across algorithms, systems, and frameworks [164]; and the method families, Fast/Stable perspectives, and practical comparisons developed in this survey. The three views emphasize different aspects of generation. Author-created schematic.

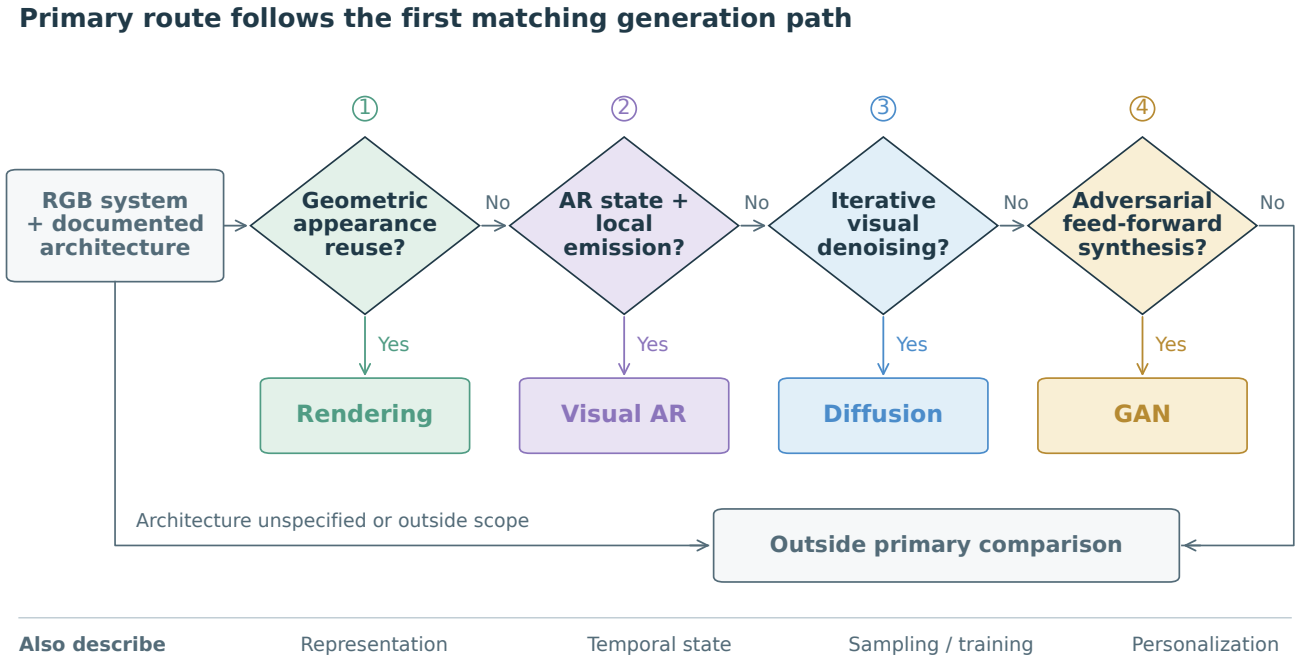



**Fig. 4** Ordered assignment to four visual-generation routes. The first satisfied rule identifies the primary route; auxiliary and temporal roles are recorded separately in Table 2. Unsupported assignments are excluded. AR: autoregressive; VAE: variational autoencoder.

stability, a review of evaluation practice, and a stage-resolved study of open implementations. Section 7 develops four observations:

1. **Decoder choice can change the speed–quality trade-off.** PersonaLive's within-source ablation quantifies this effect; it does not isolate the runtime bottleneck.
2. **Throughput gains need not shorten startup.** Live Avatar's serving configurations illustrate different effects of call reduction, extra GPUs, decoder parallelism, and bundled kernel optimizations.
3. **Memory management and demonstrated stability are separate results.** A fixed cache gives a resource bound; history and anchor ablations test output failures over specified horizons.
4. **Appearance reuse leaves other failure paths active.** The architectural identity anchor does not remove motion, tracking, deformation, or composition failures; DyStream and GaussianHeadTalk supply partial empirical examples.

We also measure 56 single-H200 configurations covering 55 methods: 11 GAN, 20 diffusion, 24 rendering, and one motion controller with FLAME visualization. Three measured calls follow three complete same-interpreter warm-ups. Stage times and warmed memory expose costs beyond generation (Section 8.7). Fifteen configurations retain models or a GPU worker; 41 reload each call. These heterogeneous workloads include neither native visual AR nor a matched quality assessment.

### 1.6 Organization

Section 2 defines the task, route ownership, and the cost and measurement language. Sections 3–6 review the four routes under common Fast and Stable lenses, each ending with route-specific design guidance. Section 7 analyzes cross-route trade-offs. Section 8 examines datasets, metrics, and runtime and temporal evaluation. Section 9 develops cross-route design guidance and future directions.

## 2 Preliminaries and Common Evaluation Language

We define the generation boundary, ownership rules, and common measures of efficiency and stability.

### 2.1 Problem Formulation and System Boundary

Let $V = (I_1, \ldots, I_T)$ denote the generated human video. A system may use enrollment data $e$ (e.g., subject video,

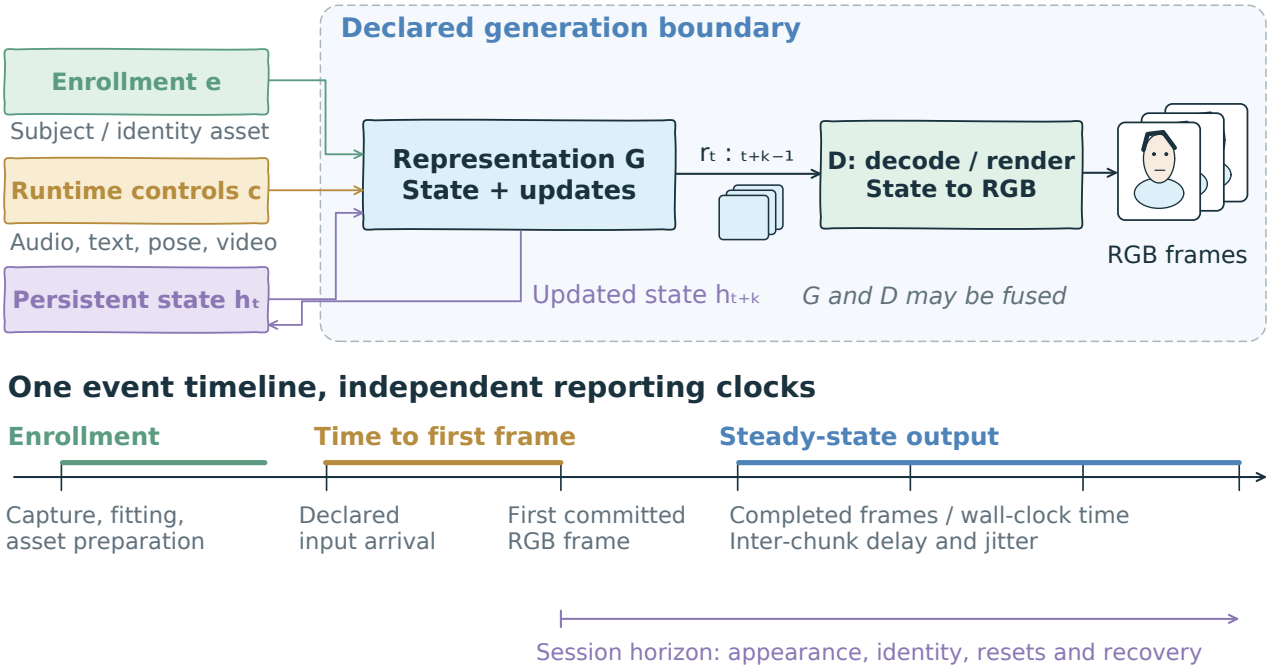



**Fig. 5** Analytical generation boundary and independent reporting clocks. The representation owner G updates visual state; decoder or renderer D produces observable RGB frames. Enrollment, time to first frame, steady-state output, and the evaluated session horizon retain their own measurement boundaries. G and D need not be separate networks.

calibrated views, or a pre-built avatar), runtime conditions $c_{1:T}$ (e.g., audio, text, pose, expression, or a driving video), and persistent deployment state $h_t$. We write

$$(I_{t:t+k-1}, h_{t+k}) = \mathcal{G}(e, c_{\mathcal{V}_t}, h_t; \theta), \tag{1}$$

where $k$ is the emitted frame or chunk length and $\mathcal{V}_t$ is the portion of the conditioning stream consumed at emission time. This formulation covers one-shot, subject-specific, and generalizable systems without assuming that a reference image is supplied at every runtime step.

Three requirements recur: *identity fidelity* preserves the subject, *driving fidelity* follows the requested speech and motion, and *temporal fidelity* maintains coherence across outputs. Appropriate metrics depend on anatomy, modality, personalization, and paired-reference availability.

For analysis, we expose an intermediate representation $r$ and a decoding or rendering operation $D$:

$$\begin{aligned}(r_{t:t+k-1}, h_{t+k}) &= G(e, c_{\mathcal{V}_t}, h_t),\\ I_{t:t+k-1} &= D(r_{t:t+k-1}).\end{aligned} \tag{2}$$

Equation (2) separates representation generation from visible-output production. The operations may be fused, and enrollment may be compiled into the decoder's parameters; subject information needed at decoding is included in $r$ or those parameters. This accounting distinguishes repeated inference from enrollment cost.

Figure 5 identifies the representation/state owner $G$, the RGB decoder or renderer $D$, and separate clocks for enrollment, first output, throughput, and session duration.

### 2.2 Efficiency and Stability

Efficiency measures the declared conditioning, synthesis, decoding, and system pipeline. Stability measures changes in identity, appearance, synchronization, and motion over a declared horizon, including state updates and recovery.

Duration claims, qualitative demonstrations, and time-resolved measurements provide different evidence. A bounded cache limits storage; generated output establishes drift. Sequential emission and responsiveness describe operating capabilities. Enrollment time remains separate from generation duration.

### 2.3 Ordered Route Ownership

For systems in Section 1.1, we inspect the documented representation, temporal update, and path to RGB, then assign the first matching rule (Figure 4):

1. **Render-based.** A subject-specific explicit or neural scene, including a mesh, neural field, Gaus-

sian set, or identity-conditioned warp, owns appearance and is queried or updated before rendering. A learned model that predicts compact motion, geometry, deformation, or Gaussian attributes remains a controller.

2. **Autoregressive.** Otherwise, a native autoregressive model owns cross-frame dense or discrete visual state, such as visual tokens or visual latents. A downstream diffusion process or renderer may be a local emission head, but it does not own the evolving visual sequence.
3. **Diffusion-based.** Otherwise, diffusion or flow matching owns dense pixels, dense visual latents, or sequential dense visual chunks. A generic variational autoencoder (VAE) decoder does not displace this ownership.
4. **GAN-based.** Otherwise, an adversarially trained feed-forward generator owns final dense or regional visual synthesis. Adversarial loss attached only as a regularizer is insufficient.

Assignments require a documented path matching one of these rules. The taxonomy organizes the reviewed families rather than exhaustively partitioning video synthesis.

A diffusion motion controller can belong to a render-based system, and an autoregressive dialogue or pose model need not own visual state. Auxiliary roles and missing performance fields are recorded independently.

*An operational test for hybrid systems.* Primary routes index the visual-generation path; representation, temporal factorization, training/sampling, and personalization are annotated separately. The visual owner need not be the latency bottleneck or source of drift.

For the AR–diffusion boundary, inspect where the noisy current sample enters the inference graph. A visual AR backbone computes a next-unit state $h_t = F(c_{\le t}, z_{<t})$ independently of the local sampling noise; a head then samples $z_t$ conditioned on that fixed state. The sampled unit is committed to the next backbone step. A chunkwise diffusion backbone instead repeatedly evaluates $v_\theta(z_t^{(s)}, s, z_{<t}, c)$ while denoising the current chunk, with the current noisy sample and noise level entering the temporal visual backbone. Both constructions factor generation causally. We index the former as visual AR and the latter as diffusion, retaining local iterative sampling in the cost breakdown in either case.

For rendering, inspect whether an identity-specific spatial representation is queried or transported under geometric control before image decoding. This places StyleAvatar's rendered UV features and compositional decoder alongside LivePortrait's source-volume warping. Adversarial losses do not change that assignment. StyleAvatar remains a contextual GAN example (Section 3.4) but is counted once. Static features alone, without geometric query or transport, do not satisfy the test.

Figure 6 contrasts the entry points of noisy samples and geometric controls in six representative systems.

## 2.4 Module Roles in Hybrid Systems

Hybrid systems distribute appearance, temporal state, motion control, decoding, and composition across modules. Table 2 shows this division for ten representative systems. The primary route describes how appearance is generated, while the other modules explain how the system is controlled and served. Their costs depend on the execution schedule: a component that delays the first output may differ from the stage that limits steady-state throughput.

The route chapters discuss the corresponding computational and temporal trade-offs. Our runtime study separates preprocessing, generation, and postprocessing under the repeated-call protocol in Section 8.7, retaining each implementation's operator boundaries and model-residency mode.

## 2.5 Route-Specific Cost Structures

Each route exposes different recurring costs. We use the following shared conventions before deriving its cost decomposition.

$C$ denotes additive work in one declared unit, such as FLOPs, GPU-seconds, or serial-stage milliseconds. Overlapping stages cannot be summed into latency. For ideal stages with times $t_1, \ldots, t_m$, first output costs approximately $\sum_i t_i$ plus buffering, whereas chunk cadence is bounded below by $\max_i t_i$. Communication and contention increase these costs. Enrollment is separate or amortized over a declared frame count.

**GAN-based synthesis.** A trained GAN predicts appearance without iterative sampling [58, 89, 90]. One evaluation fixes call count; network size, resolution, preparation, composition, and hardware determine latency.

**Diffusion and flow matching.** Diffusion [69] and flow matching [53, 117] traverse a noise-to-data path. Cost depends on the generated unit, strict number of function evaluations (NFE), per-evaluation work, conditioning, codec [96, 149, 158], and scheduling. Distillation [127, 160, 174], caching, and parallelism reduce different terms.

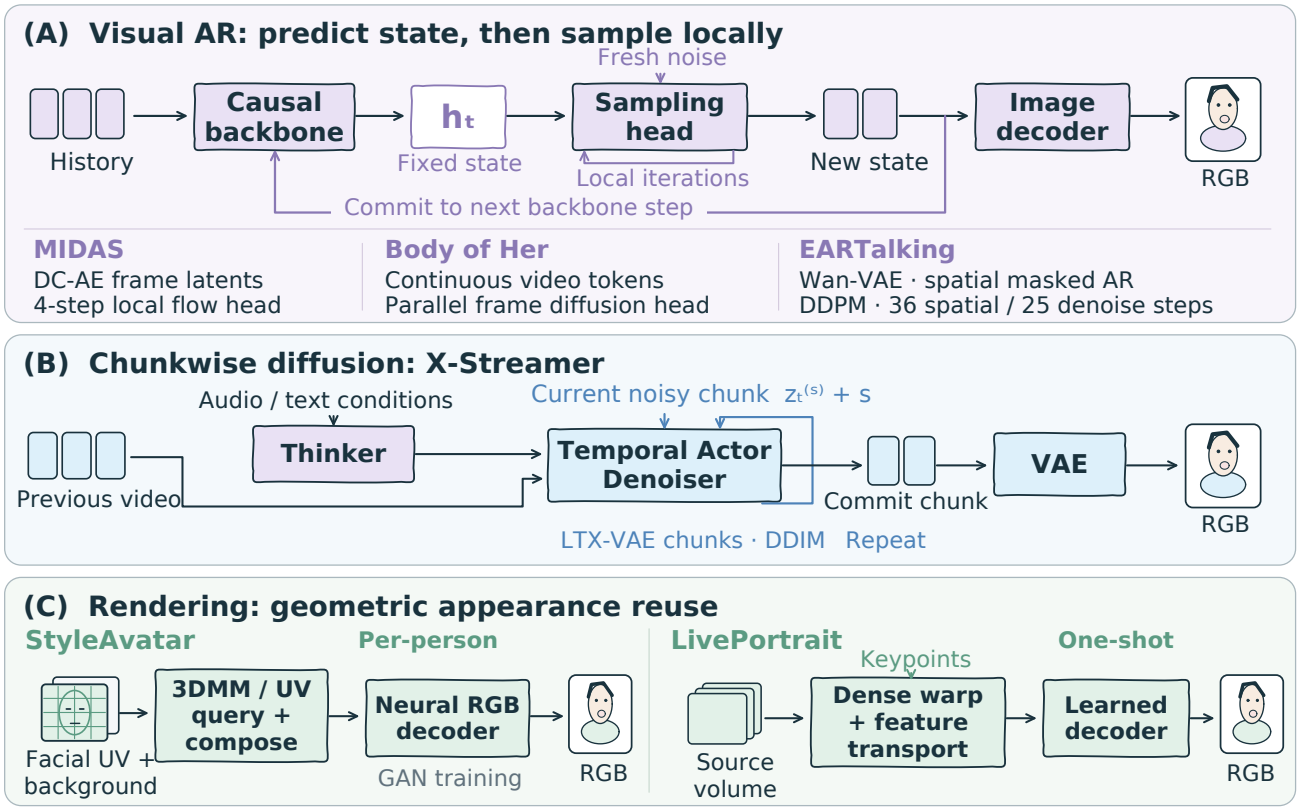



**Fig. 6** Inference paths that resolve hybrid assignments. (A) MIDAS, Body of Her, and EARTalking separate causal visual-state prediction from local sampling [5, 27, 204]. (B) X-Streamer's temporal visual backbone repeatedly consumes the current noisy sample [208]. (C) StyleAvatar and LivePortrait reuse appearance through geometric queries or feature transport before learned image decoding [62, 194]. Representation, personalization, and auxiliary modules remain separate annotations; primary route does not identify the runtime bottleneck.

**Autoregressive visual state.** Autoregression predicts discrete tokens [139] or continuous latents [24] from prior state [14, 155, 209]. Cost depends on granularity, serial depth, attention, cache updates, and decoding. Key–value (KV) caching saves repeated projections but can grow memory; windows, summaries, eviction, and resets control that growth. Parallel or speculative emission reduces serial work [102].

**Render-based synthesis.** Rendering amortizes subject-specific appearance while repeatedly applying compact controls. Costs include enrollment, tracking, control, deformation, field queries or rasterization, decoding, and composition. NeRF ray sampling and Gaussian sorting expose distinct work [91, 137, 138]. A fixed scene anchors appearance but leaves temporal failures elsewhere in the pipeline.

## 2.6 Conditions and Intermediate Representations

Runtime conditions specify speech or motion through audio, text, pose, expression, gaze, or video. Upstream tracking costs belong to the pipeline [19, 224]. Enrollment specifies appearance through a reference, monocular recording, multiview capture, or pre-built asset.

Intermediate representations determine the interface between control and appearance. Landmarks and keypoints are interpretable, but they omit appearance. Facial action units [52], three-dimensional morphable models (3DMMs), Faces Learned with an Articulated Model and Expressions (FLAME), the Skinned Multi-Person Linear Model (SMPL) and its expressive extension SMPL-X, and blendshape parameters expose compact geometric controls [109, 123, 145]. Flow fields and identity-conditioned warps transport source features. Learned motion latents trade semantics for capacity, whereas visual tokens or latents carry dense appearance state. Neural fields and Gaussian primitives encode subject-specific scenes [91, 137]. Dimensionality alone does not determine speed. Compact controls may feed an expensive renderer, while a dense latent may use a fast decoder.

**Table 2** Module roles in representative systems. Inputs list driving conditions in addition to the source identity or reference. Portrait output includes a face crop where specified.

| Method; task / input | Visual owner; route / supporting component | Temporal state and motion/control owner | Decoder / renderer; compositor |
|---|---|---|---|
| Wav2Lip+ GAN [150]; face crop / audio + source video | RGB face generator; GAN / training-only sync expert | No recursive visual state; masked source pose and audio condition each frame | Feed-forward face decoder; crop insertion is a separate output stage |
| MuseTalk [242]; lower face / audio + source | Regional latent predictor; GAN / generic VAE | Framewise visual prediction; audio conditioning and reused blend mask | VAE image decoder; mask-based blending |
| PersonaLive [113]; portrait / driving video | Dense visual diffusion; diffusion / VAE | Selected visual history and reference; motion conditions drive chunk synthesis | VAE variants |
| Live Avatar [81]; portrait / audio | Dense chunk denoiser; diffusion / causal VAE | Reference sink and rolling KV; audio-conditioned denoising | Causal VAE on a separate GPU in full system |
| MIDAS [27]; human video / multimodal controls | Continuous frame tokens; visual AR / local diffusion | AR backbone context; separate codec history; condition tokens | Four-step local head, then DC-AE |
| EARTalking [204]; portrait / audio + frame controls | Frame visual state; visual AR / local denoising | Causal context, reference sink and cyclic positions; frame-level control | Local sampling and 3D-VAE |
| StyleAvatar [194]; portrait / tracked motion | Enrolled facial UV features; rendering / adversarial decode | Fixed subject appearance; 3DMM tracking drives per-frame geometry | OpenGL UV rendering and StyleUNets; learned regional masks |
| LivePortrait [62]; portrait / driving video | Source feature volume; rendering / learned decode | Source identity and keypoint-driven deformation; motion extractor | Feature warping and neural decoder; paste-back stage |
| DyStream [22]; dyadic portrait / audiovisual input | Reused appearance volume; rendering / motion generation | Motion controller with anchor; RGB appearance persists outside controller | Portrait feature warping and rendering |
| OmniResponse [126]; dyadic portrait / audiovisual conversation | Pretrained portrait renderer; rendering / AR response control | Multimodal AR model predicts facial coefficients and text; TempoVoice predicts speech | Coefficient decoder followed by portrait renderer |

Anatomy, personalization, representation owner, decoder/renderer, and evaluation boundary remain independent comparison axes.

## 2.7 Common Acceleration and Measurement Terms

Generic acceleration acts on five cost terms [164]:

- **Fewer evaluations:** distillation, consistency objectives, solver reduction, block prediction, or speculative emission reduce strict NFE or serial calls [102, 127, 174, 227].
- **Cheaper evaluations:** representation compression, sparse computation, lightweight backbones, quantization, and optimized kernels reduce the cost of each call. For attention, I/O-aware FlashAttention reorganizes exact-attention computation in 16-bit floating point (FP16) or bfloat16 (BF16) to reduce high-bandwidth-memory traffic [44, 45], whereas FlashAttention-3 also provides a Hopper 8-bit floating-point (FP8) path whose speed lever is lower-precision Tensor Core execution and whose block quantization and incoherent processing control quantization error [162]. These standalone kernel results do not establish an end-to-end virtual-human speedup.
- **Reuse:** feature, condition, reference, or KV caches reuse a precisely named object across denoising steps, frames, chunks, or autoregressive history [118, 129, 132].
- **Decoder/renderer reduction:** efficient codecs, bounded ray samples, primitive pruning, rasterizer design, and regional synthesis reduce visible-output cost.
- **System optimization:** compilation, mixed precision, memory layout, I/O, pipelining, and parallel execution change the measured pipeline boundary.

*Throughput* counts completed output per wall-clock time under a declared operating condition. *Frames per second (FPS)* measures frame throughput at a stated resolution and boundary, distinct from content frame rate. *Real-time factor (RTF)* divides wall time by media duration; values below one exceed playback speed. *First-output latency* starts at declared input arrival and

ends at the first committed output; *time to first frame (TTFF)* requires that output to be an observable frame. *Inter-chunk latency* measures successive committed outputs under the stated buffering policy. *NFE* counts denoiser or vector-field evaluations per generated unit.

*Real-time* means meeting playback or application deadlines under the declared load. *Streaming* commits output incrementally. *Causal* generation restricts accessible conditions and history; input look-ahead, buffering, and the causal unit must be stated. A chunk-causal backbone may still consume future audio within a chunk. *Long-horizon* denotes a duration beyond the training window or ordinary evaluation clip, both of which should be reported. These properties do not imply one another.

Reported numbers require hardware, device count, resolution, precision, batch/concurrency, warm-up, timer endpoints, and included components. Unreported settings or durations remain NR.

For $N$ frames produced in $w$ seconds at media rate $f$, FPS $= N/w$ and RTF $= wf/N = f/\text{FPS}$ when workload and boundary match. Neither gives TTFF: a 12-frame chunk generated in 0.30 s achieves 40 FPS, but collecting its 0.48 s of audio at 25 FPS adds buffering delay. This example is illustrative.

### 2.8 Failure Modes and Stability Mechanisms

Stability preserves intended attributes while allowing requested changes. Observable failures include *appearance drift* (identity, color, or detail), *temporal discontinuity* (jitter or boundary jumps), *control failure* (desynchronization or suppressed motion), *geometric failure* (tracking, deformation, or composition), and *service failure* (stalls, memory growth, or resets). Recursive visual feedback is only one cause: fixed appearance can coexist with tracker jitter.

Table 3 separates seven measurement dimensions without combining them into one score. Failures may originate in tracking, control, composition, or serving, independently of the visual owner.

Five operations recur in streaming generation, especially diffusion. Their labels describe mechanisms rather than guarantees.

**L1: Persistent identity anchor.** Retain a source image, appearance asset, or reference latent across updates. Test both identity preservation and following intended changes, which a fixed anchor can resist.

**L2: Bounded multiscale temporal memory.** Preserve recent state and selected or compressed older temporal state under a documented capacity and update policy. We reserve L2 for this combination. A rolling KV window, a codec feature cache, and a learned detail-retrieval bank serve different purposes. PersonaLive selects historical frames but does not specify the memory capacity or eviction rule [113]. SoulX-LiveAct specifies fixed reference, long-term, and short-term slots; InteractiveAvatar additionally specifies promotion and replacement in its bounded long-term bank [173, 255].

**L3: Position re-indexing.** Reassign positions as memory advances, using pre-RoPE keys or cyclic indices. This limits positional extrapolation but cannot recover evicted information. EARTalking and Live Avatar use different cyclic/Rolling RoPE policies [81, 204].

**L4: History-robust training.** We distinguish *L4-G*, training on model-generated histories, from *L4-C*, corrupting ground-truth histories with noise or dropped patches. Self-Forcing exemplifies L4-G; MIDAS, EARTalking, and Hallo2 provide L4-C examples [27, 41, 80, 204]. The former reduces a training–inference mismatch over the training rollout distribution. Neither subtype establishes distribution matching for arbitrarily long or differently conditioned deployment.

**L5: Explicit boundary-state transfer.** Pass a designated motion latent, overlap state, or boundary representation from one chunk to the next. Motion Condition Injection in JoyStreamer-Flash and Avatar V's motion propagation illustrate the operation [105, 114]. Ordinary attention to previous frames is insufficient for an L5 annotation. Transfer can improve continuity or propagate an error; its direction and magnitude are empirical questions.

Evidence depends on duration and conditions. EARTalking ablates anchors and history corruption and demonstrates three training-window lengths; Live Avatar evaluates sampled windows in extended generation with repeated audio [81, 204]. These tests address component effects and temporal evolution, respectively.

Aggregate scores and selected frames can miss late degradation or interruptions. Section 8.8 describes continuous-session evaluation.

## 3 GAN-Based Methods

Adversarially trained feed-forward generators synthesize a frame or region in one evaluation [58, 89, 90]. We review regional synthesis, compact control, and appearance reuse, followed by their temporal limitations. Runtime includes conditioning and output processing as well as the generator.

**Table 3** Dimensions of temporal stability and the components that can affect them. Interpret measurements together with generation duration, input conditions, and reset behavior. Low variation alone is insufficient if the output has poor quality or fails to follow intended changes.

| Dimension | Observable and measurement | Candidate responsible components | Interpretation limit |
|---|---|---|---|
| Identity preservation | Reference-aligned identity similarity over time; failed-detection fraction | Appearance representation, visual generator, identity conditioning | A consistently wrong identity can have little drift; pose changes affect recognition. |
| Appearance preservation | Region-specific clothing/background/color or detail consistency; artifact ratings | Generator, decoder, appearance asset, compositor | Mask intended lighting, clothing, or scene changes according to the test condition. |
| Motion continuity | Pose/landmark trajectories, frame/chunk jumps, target error where paired | Motion controller, state update, tracker | Low movement may indicate frozen or suppressed motion. |
| Audiovisual synchronization | Fixed-window lip-sync measures over time; shift/error distribution | Audio encoder, motion control, audiovisual scheduling | Synchronization does not establish identity or expression fidelity. |
| Geometry/rendering stability | Wobble, disocclusion, deformation and seam errors in a declared coordinate frame | Tracking, deformation, scene representation, renderer, compositor | Static geometry or a fixed identity asset does not exclude visible artifacts. |
| State/error accumulation | Error trajectories under continuous rollout; matched state/anchor/history ablations | Visual or controller memory, eviction, boundary transfer, training history | A trend alone cannot identify its cause; bounded memory is a resource result. |
| Service reliability | Completion/failure rate, stalls, queue growth, resets, recovery time, memory trajectory | Scheduler, cache allocation, decoding and I/O | High average FPS can coexist with missed deadlines or repeated resets. |

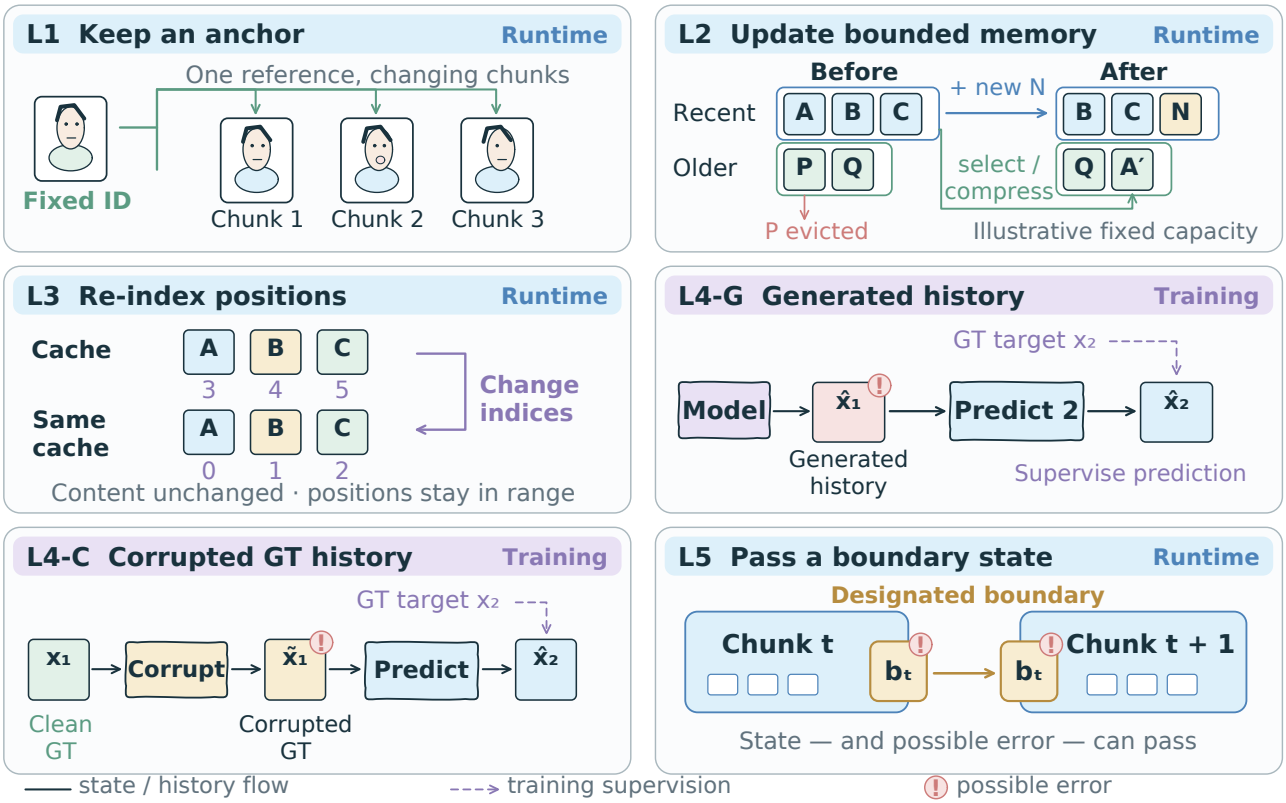



**Fig. 7** State operations behind the stability vocabulary. Runtime memory updates and explicit boundary transfer are separate from the two training constructions. A memory budget bounds storage, whereas output measurements establish whether the retained state is useful.

### 3.1 Scope, Ownership, and Cost Boundary

GAN-route assignment requires an adversarially trained generator to own dense or regional appearance after the preceding ownership tests (Section 2.3). Training-only discriminators and adversarial regularization of a warper, field, or diffusion decoder do not determine the route.

The training objective and inference path must both be checked when assigning a method. Wav2Lip reports a base generator without the visual-quality adversarial term and a Wav2Lip+GAN variant with a separate quality discriminator [150]; only the latter is included in the GAN comparison. MuseTalk reuses a pretrained variational autoencoder (VAE) and a U-Net architecture associated with latent diffusion, but performs one-call regional prediction with reconstruction, synchronization, and face/lip adversarial losses [242]. It has no diffusion objective or iterative sampler. Thus, architectural ancestry alone does not determine the route.

For a feed-forward visual path, a useful first-order deployment decomposition is

$$C_{\text{frame}} = C_{\text{condition}} + C_{\text{generator}} + C_{\text{decode/compose}} + C_{\text{system}}. \tag{3}$$

The terms cover conditioning; pixel or feature generation; decoding, blending, refinement, and super-resolution; and data movement, scheduling, and I/O. Each mechanism below reduces a different term, with system speed measured at the declared boundary.

### 3.2 Regional Synthesis: Shrinking $C_{\text{generator}}$

Regional synthesis reduces the amount of appearance regenerated per frame. Wav2Lip predicts a face crop from identity, a masked pose prior, and speech; a frozen synchronization expert supplies a training loss rather than recurrent inference state [150]. Its base and adversarial variants expose a synchronization–fidelity trade-off. Retaining the surrounding canvas also underlies VideoReTalking's editing pipeline [28], DINet's deformation-inpainting [245], landmark and appearance priors [258], memory-based mouth retrieval [144, 180], and style-space synchronization [60, 61, 94].

MuseTalk further compresses regional synthesis into a pretrained VAE latent space: a multimodal U-Net predicts the lower-face latent in one call, followed by mask-based composition [242]. Its reported throughput assumes preloaded inputs (Table 4); supplementary examples show occasional framewise jitter, motivating the temporal discussion in Section 3.6.

Training-only critics improve a generator without adding inference calls. TalkLip uses a frozen AV-HuBERT-derived lip-reading expert and improves word intelligibility over Wav2Lip on LRW/LRS2 [191]. Synchronization and quality critics likewise add supervision while leaving their own computation outside deployment.

### 3.3 Structured Control and Codebooks: Shrinking $C_{\text{condition}}$ and Bounding the Output Space

A second family separates identity from compact motion controls. MakeItTalk splits content and speaker-aware landmark motion [261], while PC-AVS separates pose, content, and identity in latent space [259]. EDTalk factorizes mouth, pose, and expression into orthogonal basis banks; its shared adversarial image generator produces RGB, and a normalizing flow samples compact pose-bank weights [178]. Emotional adaptation provides another use of disentangled control [57].

StyleGAN-based reenactment instead controls a pretrained generator by editing latent directions, masking style channels, or refining inversions [11–13, 21]. Variants address continuous emotion [7], semantic expression [143], cross-domain heads [88], and text-guided editing [17]. These methods make the control interface compact, but do not necessarily make control extraction inexpensive: inversion and feature estimation can add work, and the full image generator remains on the inference path.

Codebooks represent generated appearance through a learned set of reusable features. FlowVQTalker combines autoregressive normalizing-flow controllers for compact 3DMM expression and pose coefficients with an adversarially trained vector-quantized image generator, which decodes a spatial code sequence into a $512 \times 512$ face [179]. A grid-based codebook provides a related efficiency-oriented representation for person-specific reenactment [193].

Retrieval reuses detail compatible with the source identity. FixTalk compresses detail features into a retrieval token and aligns the retrieved features before generation; its Enhanced Motion Indicator suppresses identity leakage from the driver [177]. The reported throughput describes the combined pipeline without isolating the contribution of retrieval or compression.

### 3.4 Personalized Compositional Generators: Amortizing Appearance

Subject-specific training trades enrollment cost for repeated appearance reuse. Live Speech Portraits maps

streaming audio through predicted motion to personalized adversarial synthesis [124]. StyleAvatar combines precomputed facial-UV and background features with dynamic StyleUNets and neural-texture composition [194]. Its static features reduce recurring computation at the cost of subject-specific training; geometric appearance reuse places this hybrid in Table 14 under the render-first rule. RAM-Avatar extends personalized adversarial synthesis to full-body control [48].

### 3.5 Fast Design Guidance

Table 4 summarizes representative methods. Four levers recur, each acting on a different term of Equation (3):

1. **Regional synthesis** ($C_{\text{generator}}\downarrow$). Crop and latent-space prediction reduce generated content; codec reconstruction, blending, seams, and mouth interiors still require evaluation.
2. **Compact controls** ($C_{\text{condition}}\downarrow$). Basis banks, coefficients, latent directions, and codebook indices replace dense features, but restrict expressiveness to their control space.
3. **Appearance reuse** ($C_{\text{generator}}\downarrow$). Static features trade enrollment for repeated savings; reference-conditioned retrieval can support one-shot use. Preparation and generalization costs differ.
4. **Fixed-path optimization** ($C_{\text{system}}\downarrow$). Compilation, reduced precision, and fused kernels reduce implementation overhead. StyleAvatar's FP16 TensorRT configuration illustrates hardware-dependent gains.

These mechanisms can be combined, but preparation and I/O may dominate once synthesis is cheap. Tables 4 and 14 retain source-specific operating points.

### 3.6 Stable Design Guidance

Framewise generation avoids recursive visual error when previous outputs are not reused, but does not enforce consistency. MuseTalk reports jitter; changing pose estimates or reference features can also destabilize output. Recurrent controllers and retrieval memories introduce separate state errors.

Using Section 2.8, fixed references and feature banks provide L1 anchors. FixTalk retrieval and FlowVQTalker coefficient history do not establish bounded multiscale memory (L2); L3 applies only to positional sequential components. L4 history training and L5 boundary transfer remain less developed here. The sources in Table 4 lack joint minute-scale, time-binned identity, synchronization, motion, and quality evaluation.

### 3.7 Open-Source Measurements on One H200

Table 5 complements the paper reports with 11 configurations, each measured three times after three same-interpreter warm-ups. Ten retain audited model objects; MakeItTalk reloads each call. Output dimensions describe the visible region or composited canvas, which can exceed the internally synthesized face crop. MuseTalk and VideoReTalking now have completed re-measurements.

The measurements show substantial work around the feed-forward generator. MuseTalk averages 0.85 s in Gen for 200 frames, while preparation and output processing take 20.68 and 19.99 s; its complete call takes $41.53 \pm 0.06$ s (sample standard deviation). DINet's preparation exceeds neural generation even with supplied landmarks, and PC-AVS spends 11.94 s on restoration/output processing versus 2.44 s in Gen. These within-configuration observations motivate profiling the surrounding pipeline; they do not establish streaming latency or reproduce source claims made with preloaded inputs.

### 3.8 Synthesis and Open Questions

Regional synthesis requires boundary-quality tests, compact controls require coverage and extraction-cost tests, and personalization requires enrollment accounting. Matched long-session evaluation is still needed to distinguish reduced recursive drift from framewise jitter.

GAN efficiency follows from feed-forward prediction, regional output, compact control, and appearance reuse. Temporal behavior also depends on conditioning, so generator speed and full-pipeline stability remain separate questions.

## 4 Diffusion-Based Methods

Diffusion and flow models condition video on identity, speech, text, pose, and history [69,73,116,117,183]. Acceleration reduces sampling calls, representation size, or repeated computation. Streaming adds the need to remain consistent under self-generated history. We review these mechanisms through their computational costs and state updates.

### 4.1 Scope and Cost Model

A denoising diffusion probabilistic model (DDPM) repeatedly invokes its denoiser [69]. Flow matching avoids

**Table 4** Representative GAN-based visual generators. Timing values retain the source paper's hardware, resolution, and component boundary and are not directly comparable. NR denotes not reported in the primary paper.

| Method | Visual owner / scope | Primary Fast mechanism | Source timing boundary | Stable boundary |
|---|---|---|---|---|
| Wav2Lip+ GAN [150] | Feed-forward RGB face-crop generator; identity-general | Face-crop rather than full-canvas synthesis; training-only sync and quality critics | FPS, hardware, resolution, TTFF, and timer NR | Stateless framewise generation; evaluated duration NR |
| MuseTalk [242] | U-Net-generated lower-face VAE latent; no subject adaptation | One-call regional latent prediction; reusable blend mask | 30 FPS, $256^2$, one V100, preloaded data; included stages NR | Single-frame jitter reported; quantitative long-horizon evaluation NR |
| TalkLip [191] | Feed-forward $96^2$ RGB face-crop generator; identity-general | Training-only lip-reading expert improves intelligibility | FPS, latency, hardware, parameters, and timer NR | Stateless framewise generation; long-horizon evaluation NR |
| Live Speech Portraits [124] | Person-specific adversarial portrait generator | Compact predicted motion; person-specific appearance | Source-reported 30 FPS system rate on a consumer GPU; timer boundary NR | Recurrent motion prediction; long-horizon evaluation NR |
| EDTalk [178] | Adversarial RGB generator; mouth, pose, and expression banks; one-shot | Compact reusable bases and progressive freezing | Inference timing NR; about 7 h disentanglement training on two 3090 GPUs | Compact controller state only; long-horizon evaluation NR |
| FlowVQ Talker [179] | Adversarial VQ image generator; full $512^2$ portrait | Compact coefficient control and learned spatial codebook | Inference FPS, latency, hardware, memory, and timer NR | Coefficient history in the controller; long-horizon evaluation NR |
| FixTalk [177] | Adversarial dense portrait generator; one-shot | Identity-suppressing EMI and compressed memory-retrieved EDI detail | 27.601 FPS on A100 80 GB; $512^2$ data, full timer boundary NR | Learned detail retrieval (not temporal L2); TTFF and long-horizon evaluation NR |

EDI: Enhanced Detail Indicator; EMI: Enhanced Motion Indicator. Timing values are taken from the cited papers.

**Table 5** Our single-H200 measurements of 11 GAN-route configurations after three complete warm-ups. Time in seconds; sampled memory in GiB.

| Configuration | $N$ | $W \times H$ | Pre | Gen | Post | Load | Call $\pm s$ | Warm peak $M_w$ (GiB) |
|---|---|---|---|---|---|---|---|---|
| Wav2Lip+GAN$^R$ [150] | 248 | 256×256 | 0.12 | 0.09 | 0.57 | $< 0.01$ | $0.78 \pm 0.01$ | 3.87 |
| MuseTalk$^R$ [242] | 200 | 704×1216 | 20.68 | 0.85 | 19.99 | 0.01 | $41.53 \pm 0.06$ | 9.49 |
| TalkLip$^R$ [191] | 251 | 700×700 | 2.63 | 0.03 | 2.22 | $< 0.01$ | $4.93 \pm 0.02$ | 14.54 |
| Live Speech Portraits$^R$ [124] | 672 | 512×512 | 0.99 | 7.41 | 10.47 | $< 0.01$ | $18.88 \pm 0.22$ | 2.08 |
| EDTalk$^R$ [178] | 251 | 256×256 | 0.92 | 4.64 | 1.70 | $< 0.01$ | $7.27 \pm 0.02$ | 3.82 |
| VideoReTalking$^R$ [28] | 147 | 700×700 | 19.01 | 10.17 | 31.40 | $< 0.01$ | $60.60 \pm 0.16$ | 16.01 |
| DINet$^R$ [245] | 241 | 1920×1080 | 40.64 | 2.36 | 3.79 | $< 0.01$ | $46.79 \pm 0.11$ | 1.85 |
| MakeItTalk [261] | 287 | 256×256 | 2.72 | 2.46 | 2.39 | 2.43 | $10.00 \pm 1.25$ | 2.79 |
| PC-AVS$^R$ [259] | 361 | 224×224 | 1.34 | 2.44 | 11.94 | $< 0.01$ | $15.75 \pm 0.15$ | 6.64 |
| EAT$^R$ [57] | 88 | 256×256 | 0.04 | 2.01 | 0.51 | NR | $2.56 \pm 0.01$ | 2.95 |
| HyperReenact$^R$ [11] | 101 | 256×256 | 7.15 | 9.12 | 0.18 | NR | $16.48 \pm 0.21$ | 5.96 |

Means over three measured calls after three complete same-interpreter warm-ups; Call includes sample standard deviation. $R$: audited models/worker remain resident; other configurations reload each call. Load NR means no recorded interval, not zero. Call spans inference entry to synchronized return, including loading and native validation, excluding generic output QA. Warm peak $M_w$: maximum NVML peak over measured calls, sampled approximately every 100 ms across preparation, generation and output processing. $N$: output frames; dimensions exclude comparison panels but may exceed synthesis crops. Workloads differ. Full timing definitions and records: Section 8.7.

Ten configurations retain audited models; MakeItTalk reloads each call. DINet and EAT accept supplied preprocessing products. See Table 24.

**Table 6** GAN-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

| Target | Design options | Trade-offs to check |
|---|---|---|
| Low per-frame latency, general identity | regional synthesis; one-call generator | Seams, mouth-interior quality; preparation can dominate measured runs |
| High throughput at high resolution | compact controls and retrieval; Potential: transfer compiled FP16 serving | Control space bounds expressiveness; hardware-specific |
| Repeated use of one subject | person-specific appearance reuse | Hours-scale enrollment; no zero-shot use |
| Minute-scale stability | reference and temporal control; Potential: long-session deployment | Framewise jitter and controller drift; joint long-horizon evaluation NR in the reviewed systems |

ODE simulation when constructing training targets, but inference still integrates a learned vector field [53, 117]. In a video diffusion Transformer (DiT), token count, network dimensions, and attention extent govern each call [146, 223]. The deployed cost is approximately

$$C_{\text{gen}} = N_{\text{FE}} C_{\text{denoise}} + C_{\text{condition}} + C_{\text{decode/render}} + C_{\text{system}}. \quad (4)$$

NFE counts solver stages, classifier-free guidance (CFG) branches, and extra cache-refresh calls. The remaining terms cover condition encoders; VAE decoding, super-resolution, or rendering; and scheduling, communication, buffering, and I/O. As NFE decreases, the bottleneck can shift to a DiT call, codec, or condition encoder. Component gains therefore do not multiply directly into system speedups.

Two service measures complement throughput. TTFF includes input acquisition, encoding, first-chunk generation, decoding, and buffering. Inter-chunk latency and jitter determine whether playback continues without stalls. Audio acquisition therefore imposes a latency floor even with a fast denoiser. We retain the source boundaries when comparing these quantities.

Dense visual diffusion generates pixels or video latents through a noise-conditioned visual backbone; a codec then reconstructs RGB. FADA, RAP, StreamAvatar, SoulX-FlashTalk, and Avatar V illustrate this path [51, 114, 165, 176, 257]. Compact motion generators followed by identity-conditioned warping or scene rendering belong at the render boundary, including IF-MDM, INFP, AudioRTA, Avatar Forcing, EmbodiedHead, FloodDiffusion, and OmniTalker [16, 93, 100, 203, 220, 243, 265]. Figure 6 resolves the harder distinction between an iterative temporal backbone and a fixed causal visual state with a local iterative head.

Normalizing flows use invertible transformations and tractable Jacobians [142]. Flow matching instead learns a noise-to-data velocity field, sampled through ODE integration or a distilled approximation [117]. Its simulation-free training objective does not imply one-call inference. Strict NFE must include solver stages, guidance branches, and auxiliary calls.

Many mechanisms exploit a task prior: appearance changes more slowly than speech and motion. Identity, clothing, and background can be cached or anchored, while mouth shape and expression must follow current inputs. READ aligns compressed video with SpeechAE [189]; RAP uses hybrid audio attention [51]; and SyncCache recomputes light audio blocks while reusing heavy DiT residuals within a reverse trajectory [128]. Long-video systems similarly retain appearance anchors and recent motion while compressing older history.

Equation (4) locates the main levers: distillation reduces NFE; compressed latents, sparse attention, and low precision reduce per-call work; caches remove repeated conditioning; codecs reduce decoding; and parallel pipelines reorganize execution. Systems combine these levers, so measurements must identify their active components [81, 128, 257].

Figure 8 separates these runtime optimizations from state operations defined in Section 2.8.

### 4.2 Fast I: Compact Representations and Efficient Backbones

Representation compression follows two paths. Compact motion or expression diffusion delegates appearance to a decoder or renderer, reducing iterative state but limiting the control space. Dense visual diffusion instead compresses video through a spatiotemporal VAE, retaining broader appearance synthesis while paying for audio alignment, detail recovery, and decoding. Their FPS values cover different work.

Compact controllers move stochastic generation out of dense visual space. IF-MDM and INFP generate facial or conversational motion latents; AudioRTA distills expression prediction before a mesh/Gaussian decoder; Avatar Forcing predicts reactive dyadic motion; and EmbodiedHead generates state-conditioned FLAME controls [93, 100, 220, 243, 265]. FloodDiffusion studies changing streaming conditions in skeleton-motion space [16]. Their small iterative state reduces

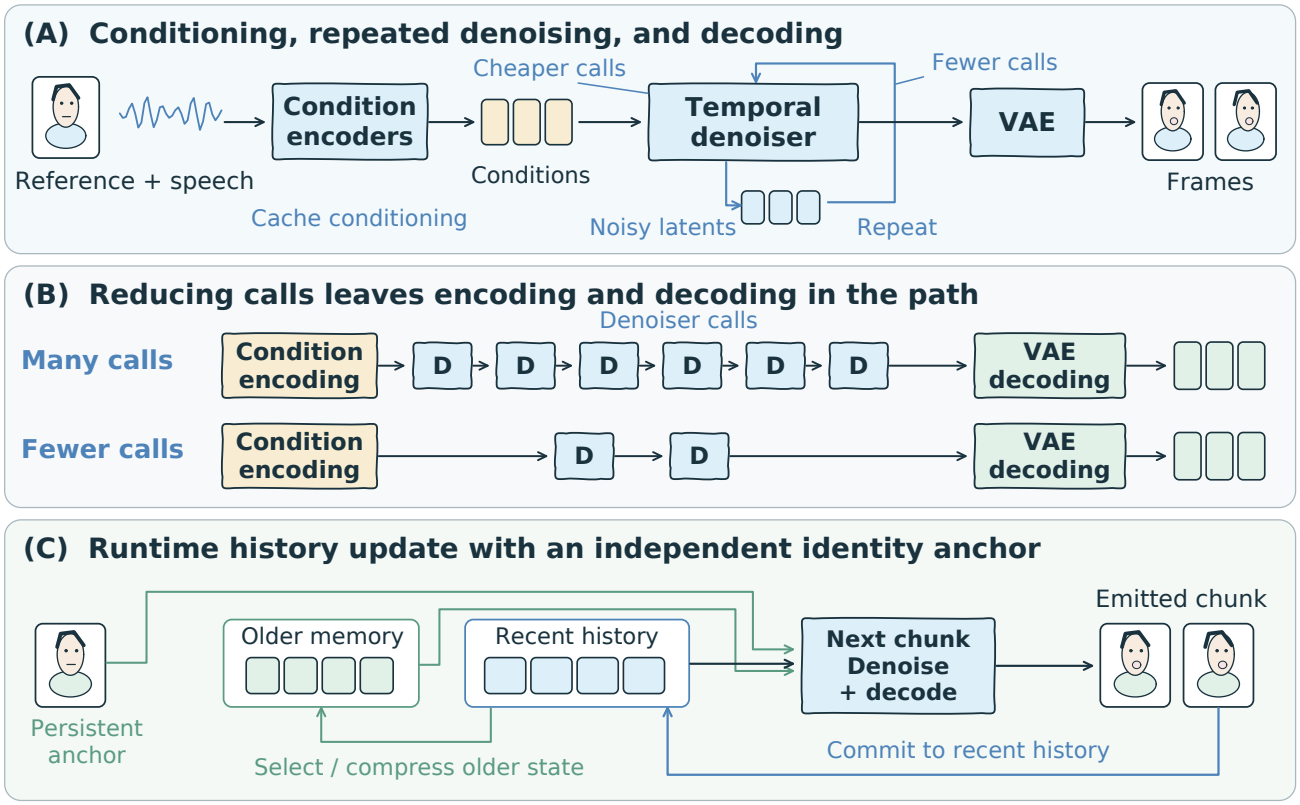



**Fig. 8** Diffusion generation as a runtime path and a state-update process. (A) Acceleration acts on distinct components. (B) After denoiser-call reduction, other stages still contribute to runtime; Section 7 gives a measured decoder trade-off. (C) The committed chunk updates recent history, while older state is selected or compressed under the declared policy. Block sizes are schematic and do not represent measured stage durations. Training constructions are shown separately in Figure 7.

controller work, but appearance and its rendering cost remain downstream (Section 6).

A compact controller does not necessarily imply a render-based visual owner. Motar predicts 512-dimensional motion with a causal Transformer and a local diffusion head, but its downstream decoder performs four-step block-causal diffusion over dense visual content [3]. The complete RGB pipeline therefore remains in the diffusion route. This contrasts with compact controllers that transport an identity-specific appearance representation before decoding; the word "renderer" alone does not resolve the distinction.

Dense video models compress space and time. Stable Video Diffusion combines image pretraining, video curation, and temporal-layer training [10]. HunyuanVideo, Wan, and CogVideoX couple a spatiotemporal VAE, DiT, and text encoder [98, 186, 223]: compression sets token volume, attention governs per-call work, and chunked decoding permits early emission. LTX-Video co-designs its VAE and token-matched DiT [65]. For virtual humans, compression must preserve teeth, lip boundaries, eyelids, and rapid expressions.

With temporal and spatial compression factors $r_t$ and $r_s$, video-token count scales approximately as $THW/(r_t r_s^2)$ [65, 146]. Compression reduces attention, MLP, KV, and communication costs, but assigns fewer tokens to facial detail and groups more phonetic changes into each temporal token. READ therefore pairs a Temporal VAE with a time-aligned SpeechAE [189], while RAP combines compressed LTX-VAE latents with hybrid audio attention and latent inheritance [51]. Their mechanisms emphasize alignment and detail recovery alongside token reduction; reported operating points remain in Table 7.

Reference-guided decoding can recover compressed detail. Xu et al. inject reference features into a deep-compression VAE while a rectified-flow Transformer generates dense video latents [212]. This remains visual diffusion. Their timing assumes precomputed reference features, separating preparation from per-chunk decoding.

Long references create a second attention bottleneck. Avatar V uses asymmetric sparse reference attention, reuses reference K/V and rotary position embeddings across denoising steps, and adds one-step super-

resolution [114]. Its relative latency result motivates conditioning caches, but no absolute runtime is reported.

VAE decoding remains on the inference path after DiT distillation (Figure 8B). PersonaLive's TinyVAE option trades identity, motion, and temporal quality for throughput; MotionStream also uses TinyVAE in its fastest reported configuration [113, 166]. Wan instead supports chunked codec execution with cached adjacent features [186]. Compression should therefore be assessed with decoder latency and facial reconstruction quality, as well as denoiser cost.

*Conditioning cost and expression control.* Conditioning must preserve articulation and broader expression without making its own cost invisible. EmotiveTalk separates lip-related audio information from emotion and expression, then expands temporal cues through emotional video diffusion [190]. Its additional encoders and attention paths belong in the runtime breakdown, while synchronization, motion naturalness, and identity remain separate quality targets. HunyuanPortrait uses identity-disentangled implicit motion features and attention adapters to condition a video-diffusion backbone on driving-video expression and pose [214].

### 4.3 Fast II: Few-Step and One-Step Distillation

Few-step generation combines three distinct operations: trajectory rectification or consistency training connects distant noise states; distribution matching distillation (DMD) trains a short-path generator from real/generated score differences; and CFG distillation folds guidance branches into one call. These change the path, student distribution, and calls per step, respectively. Strict NFE also includes inference-time cache-refresh calls.

The generic lineage requires precise objectives. Consistency Models map points on the same probability-flow ODE trajectory to a common output; consistency distillation depends on a pretrained diffusion trajectory, whereas standalone consistency training learns directly from data [174]. Latent consistency models and their LoRA form made few-step sampling broadly practical for latent image diffusion [127]. DMD estimates a Kullback–Leibler gradient direction from the difference between real-data and generated-distribution scores and adds a perceptual regression term anchoring the student to its teacher's conditional sample [227]; DMD2 addresses stale generated-score estimates and unstable optimization through two-timescale updates, adversarial supervision on real data, and backward simulation of inference-relevant states [226]. These generic image results motivate virtual-human students, but do not settle identity, synchronization, or self-history errors.

Call accounting separates deployed denoiser evaluations from training-only teachers/critics and from decoding. CFG, solver substages, and clean-latent cache updates contribute to NFE. When guidance or solver details are absent, we retain the reported step count. An NFE ratio without matched wall-clock measurements remains a call-count ratio.

FADA distinguishes trajectory shortening from guidance folding. Its six-step PeRFlow student reduces a three-branch, 25-step DDIM teacher from 75 denoiser calls to 18; replacing those branches with a learned CFG token reduces the count to six [257]. These are call-count reductions, with hardware and wall-clock runtime unreported: fewer steps need not mean one network evaluation per step.

Live Avatar shows why cache refreshes also count. Four-step DMD removes CFG, but conventional Self-Forcing adds a clean-cache evaluation. History Corrupt instead keeps historical KV at the current block's noise level, eliminating that extra call [81]. The resulting step-matched state also supports a deployment pipeline that assigns one denoising timestep to each device.

LLIA combines few-step consistency and adversarial training with TensorRT, INT8 execution, and VAE–U-Net overlap [230]. TurboTalk progressively distills a four-step DMD student to one author-counted evaluation per chunk [120]. The former reports a deployed operating point, whereas the latter reports an NFE ratio without wall-clock measurements; call reduction and elapsed-time improvement remain distinct quantities.

The generated unit also matters. Cui et al.'s Avatar-Forcing performs one joint update per sliding-window advance; each block receives four updates in the default four-block window [43]. This differs from one evaluation per isolated block and from Ki et al.'s motion-space Avatar Forcing [93]. LeapTalk instead changes the generative path through a reference-anchored Brownian bridge and SNR-aligned time transformation [239]. Step comparisons require the unit, guidance, and update schedule.

Block-autoregressive distillation must address the mismatch between training histories and self-generated deployment state (L4-G). MotionStream's Self Forcing, StreamAvatar's and LiveTalk's on-policy rollouts, SoulX-FlashTalk's multi-block self-correction, and AsymTalker's asymmetric teacher–student histories expose students to their own errors [29, 125, 165, 166, 176]. The objective must therefore preserve the distribution of a rollout as well as the quality of an isolated clip.

Task-specific distillation protects signals that generic distribution matching can discard. Lip Forcing schedules teacher CFG by noise level and weights updates with SyncNet confidence; its deployed count includes a clean-cache call in addition to denoising [31]. Hallo-Live uses audiovisual preference evaluators in DMD [104]. Such supervision can preserve synchronization or identity while inheriting evaluator bias, so teacher–student quality and long-rollout degradation should accompany NFE and runtime.

Maintaining appearance is insufficient if distillation suppresses intended motion. DynaForcing observes diminished dynamics in its self-forcing setup and combines data-anchored rollouts, reward-weighted updates, and reference perturbation [82]. Routed Forcing further separates semantic regions and noise stages: real-data supervision targets non-mouth person regions at high noise levels, while distribution matching is retained elsewhere and during detail refinement [175]. These results identify a failure mode in the tested training regimes, not an inevitable outcome of all distribution matching. Motion amplitude and diversity should therefore accompany identity and synchronization scores when evaluating a distilled avatar.

### 4.4 Fast III: Computation Reuse and System Parallelism

Adjacent denoising steps offer reusable computation. LightningCP refreshes high-level decoder features at selected timesteps and allocates finer attention to the foreground [122]; Follow-Your-Emoji-Faster uses a timestep-aware information cache [131]. Their reported improvements concern fixed-clip denoiser timing, with incomplete hardware disclosure (Table 7). Flow caching skips reusable velocity-field computation [132], while speculative decoding drafts inexpensive continuations for model verification, extending a language-model technique to video diffusion [76, 102].

Selective reuse exploits slow appearance and rapidly changing speech. SyncCache probes portrait-weighted changes in early Transformer features and refreshes the full network when accumulated error crosses a threshold. Between refreshes, heavy DiT residuals are reused while lightweight audio blocks are recomputed [128]. Its two backbone evaluations illustrate this denoising-time trade between reuse and conditioning responsiveness.

Denoising-time and sequence-time caches preserve different state. The former reuses features across sampling steps of one chunk; the latter carries context into later chunks and can also propagate errors. Refresh frequency should follow the stored signal: identity and background can tolerate longer reuse than current speech. Avatar V caches step-invariant reference K/V; SoulX-LiveAct aligns adjacent chunks at the same diffusion step with Neighbor Forcing and compresses older history with ConvKV [114, 255]. Hybrid KV compression extends this approach [86]. Evaluation should expose refresh rules, cache depth, warm-up cost, memory, and resulting quality [81, 128, 255].

Cache lifetime determines what a reuse mechanism can preserve. Avatar-Forever's ForeverCache computes historical features at the first denoising step, reuses them within the current chunk, and resets them for the next chunk; it is a denoising-time cache rather than an L2 memory bank [108]. Its separately trained recovery adapter instead addresses accumulated rollout errors. Vidu S1's TwinCache maintains noisy and clean historical states and selects them at different denoising stages [237]. These designs distinguish feature reuse from the choice of temporal information available to the model.

Deployment determines whether component savings reach the output stream. LLIA overlaps VAE and U-Net execution; StreamAvatar overlaps DiT and VAE while separately accounting for input buffering and startup [176]. FlowAct-R1 combines few-step sampling with reduced precision, hybrid parallelism, fused kernels, and asynchronous decoding [195]. SoulX-FlashTalk pipelines speech processing, denoising, VAE decoding, and history encoding [165]. These designs require stage-level timings because target playback rate alone does not establish sustained throughput.

Live Avatar's Timestep-forcing Pipeline Parallelism partitions denoising time across devices, each with step-matched rolling KV, and uses a separate GPU for causal VAE decoding [81]. Section 7 analyzes its serving ablation and the source's differing TTFF reports. Throughput depends on the slowest active stage, whereas startup includes buffering and the first pipeline traversal.

### 4.5 Open-Source Measurements on One H200

Table 8 reports 20 single-H200 configurations covering 19 methods. It includes Wan-Animate-2, sliding-window AvatarForcing, and third-party Moore-AnimateAnyone [43, 73, 188], the SoulX implementations, and six additional diffusion implementations. Each has three measured calls; source-reported multi-device serving results remain separate.

Residency affects call cost. LiveTalk reloads each time: Load averages 90.03 s within a 94.89 s call, while Gen takes 2.11 s. Wan retains its GPU worker and averages 69.97 s per call, with final lengths of 35, 33, and

**Table 7** Acceleration mechanisms and reported runtimes of diffusion systems. Values use the settings of the cited papers and are not directly comparable. NFE-D counts denoiser calls; NR denotes not reported.

| Method | Sampling / model calls | Reported timing | Resource and boundary |
|---|---|---|---|
| FADA [257] | 6 steps; 75 → 18/6 NFE-D | 4.17/12.5× call reduction | Wall-clock time and hardware NR |
| LLIA [230] | 2/4 steps; strict NFE NR | 78 FPS; ∼140 ms response | RTX 4090D, $384^2$; different startup/steady chunks |
| RAP [51] | Strict NFE NR | 42.41 / 39.87 FPS | One A800, $512^2$; full boundary NR |
| LightningCP [122] | Steps/NFE NR | U-Net: 23.692→7.528 s/clip | One-GPU baseline; multi-A800 method |
| Follow-Your-Emoji-Faster [131] | 30 DDIM steps | U-Net: 6.28→2.36 s | 64 frames, $256^2$; device NR |
| PersonaLive [113] | 4 steps | 15.82 FPS; 0.253 s/chunk | One H100, $512^2$; default decoder |
| MotionStream [166] | 3 NFE/chunk | 29.5 FPS; 0.39 s latency | One H100, 480p, TinyVAE; general motion |
| StreamAvatar [176] | 3 steps; strict NFE NR | ∼1.20 s first output | Two H800, 928 × 704; 25 FPS is media rate |
| Hallo-Live [104] | Steps/NFE NR | 20.38 FPS; 0.94 s latency | Two H200; full boundary NR |
| Lip Forcing [31] | 2 denoising + 1 cache call | 31.58 / 15.11 FPS | One H100, $512^2$; speech preprocessing excluded |
| FlowAct-R1 [195] | 3 base NFE; refiner NR | 25 FPS; ∼1.5 s TTFF | A100 platform, 480p; device count/full boundary NR |
| Live Avatar [81] | 4 NFE | 45.2 FPS; 1.21 s TTFF | Five H100, 720 × 400; network excluded |
| SoulX-FlashTalk [165] | 4 NFE | 32 FPS; 0.87 s startup | Eight H800, 720 × 416 |
| SoulX-FlashHead [232] | Steps/NFE NR | Lite: 96; Pro: 10.81 FPS | Lite device differs across paper sections: H20 / RTX 4090 |
| SoulX-LiveAct [255] | 3 steps; strict NFE NR | 20 FPS; 0.94 s latency | Two H100/H200; latency definition NR |
| Avatar V [114] | 24 base steps + 1 SR step | ∼ 3× relative latency gain | Eight unspecified GPUs; absolute timing NR |

34 frames. We preserve these execution differences and individual output lengths.

Emission and conditioning remain visible after sampling is shortened. For 250-frame FlashHead outputs, Lite and Pro average 2.99 and 4.53 s in Gen but 6.03 and 11.91 s per complete call; their Post stages take 2.66 and 6.49 s. This identifies costs within the measured variants, without evaluating their relative visual quality. EverAnimate and TaoMate also spend substantial time on conditioning. Sampling schedules, offloading, dimensions and output lengths differ, and full sampling/decoder settings for the six additional configurations were not supplied. Appendix A records these boundaries. The short outputs do not test the long-session claims in Table 9.

Table 7 motivates comparison along a quality–speed curve with fixed task, resolution, hardware, and timer boundaries. Call reduction, component reuse, decoder choice, and parallel serving contribute distinct gains.

### 4.6 Streaming Generation: Where Fast Meets Stable

Chunkwise generation couples speed with stability: smaller chunks reduce buffering and TTFF but increase scheduling, cache-update, and decoder overhead; larger chunks improve parallel efficiency while delaying first output. Every committed chunk also becomes history for subsequent generation. Equal throughput can therefore coexist with different response and drift behavior.

Diffusion Forcing assigns independent token noise levels and controls causal denoising through a position-dependent schedule [23]. Self-Forcing instead changes training: students consume self-generated history during on-policy rollouts, addressing exposure mismatch without inherently reducing NFE [80]. CausVid distills a bidirectional teacher into a causal student with asymmetric DMD and rolling KV [228]; Causal Forcing first aligns the teacher with the deployed autoregressive architecture [263]; and Rolling Forcing jointly denoises a moving window [119]. Wan-Streamer scales interactive streaming deployment [77, 79], while Vidu S1 combines DMD, perceptual consistency, and stage-dependent noisy/clean history reuse [237].

Virtual-human systems add identity and speech state. PersonaLive advances micro-chunks through successive noise levels [113]; StreamAvatar and LiveTalk add reference sinks and generated history [29, 176]; EchoTorrent uses a reference sink and tail-only forcing [135]. SoulX-FlashTalk, SoulX-LiveAct, Lip Forcing, and FlowAct-R1 further specialize self-correction, memory, synchronization, and full-body planning [31, 165, 195, 255]. StableAvatar and InfiniteTalk instead extend offline inputs through overlapping windows, without targeting interactive service latency [184, 219].

Streaming can change the appearance reference itself. Vidu S2 supports reference switching, editing, and spatial-video workflows [236]; FlowAct-R2 combines dynamic multimodal references with behavior and inter-

**Table 8** Our single-H200 measurements of 20 diffusion-route configurations after three complete warm-ups. Time in seconds; sampled memory in GiB.

| Configuration | $N$ | $W \times H$ | Pre | Gen | Post | Load | Call $\pm s$ | Warm peak $M_w$ (GiB) |
|---|---|---|---|---|---|---|---|---|
| StableAvatar [184] | 97 | 512×512 | 2.30 | 138.55 | 3.30 | 34.92 | 179.37 ± 2.66 | 29.30 |
| InfiniteTalk [219] | 81 | 896×448 | 13.83 | 678.05 | 12.98 | 209.38 | 916.74 ± 30.07 | 45.28 |
| LiveTalk [29] | 78 | 512×512 | 0.15 | 2.11 | 2.17 | 90.03 | 94.89 ± 26.13 | 21.06 |
| Wan-Animate-2$^R$ [188] | 35/33/34 | 528×935 | 5.76 | 61.00 | 3.18 | NR | 69.97 ± 0.06 | 108.78 |
| AvatarForcing-SW [43] | 129 | 624×624 | 0.51 | 6.40 | 6.33 | 98.25 | 111.51 ± 20.67 | 23.76 |
| EverAnimate [111] | 77 | 832×480 | 159.26 | 104.53 | 5.54 | 23.24 | 292.92 ± 5.78 | 57.28 |
| LeapTalk [239] | 28 | 512×512 | 0.20 | 0.13 | 0.72 | 3.35 | 4.49 ± 0.19 | 9.81 |
| Avatar-Forever [108] | 33 | 768×512 | 7.14 | 1.64 | 1.20 | 7.91 | 17.90 ± 0.26 | 41.82 |
| TaoMate [56] | 121 | 768×512 | 19.88 | 6.22 | 3.28 | 18.94 | 48.34 ± 2.33 | 65.50 |
| Moore-AnimateAnyone [73] | 16 | 512×784 | 0.63 | 17.65 | 1.73 | 23.24 | 43.28 ± 4.65 | 14.93 |
| SoulX-LiveAct [255] | 245 | 512×512 | 10.99 | 20.92 | 7.13 | 83.08 | 123.24 ± 13.39 | 76.18 |
| SoulX-FlashTalk$^R$ [165] | 250 | 416×720 | 1.32 | 37.18 | 7.21 | NR | 45.72 ± 0.28 | 53.79 |
| SoulX-FlashHead Lite$^R$ [232] | 250 | 512×512 | 0.38 | 2.99 | 2.66 | NR | 6.03 ± 0.05 | 6.21 |
| SoulX-FlashHead Pro$^R$ [232] | 250 | 512×512 | 0.88 | 4.53 | 6.50 | NR | 11.91 ± 0.15 | 8.33 |
| Follow-Your-Emoji-Faster [131] | 16 | 512×512 | 2.44 | 3.72 | 0.36 | 34.31 | 40.86 ± 3.24 | 21.55 |
| Hallo-Live [104] | 21 | 992×512 | 0.03 | 1.02 | 1.12 | 11.20 | 13.77 ± 0.04 | 49.29 |
| LightningCP [122] | 16 | 512×512 | 4.17 | 4.57 | 0.59 | 13.31 | 22.68 ± 0.08 | 40.64 |
| Lip Forcing [31] | 9 | 512×512 | 3.56 | 1.21 | 0.58 | 197.04 | 202.42 ± 35.75 | 50.31 |
| Live Avatar [81] | 45 | 704×384 | 17.23 | 7.22 | 8.04 | 65.01 | 98.58 ± 0.25 | 74.75 |
| PersonaLive [113] | 16 | 512×512 | 0.63 | 1.06 | 1.04 | 15.37 | 18.13 ± 0.27 | 15.82 |

Means over three measured calls after three complete same-interpreter warm-ups; Call includes sample standard deviation. $R$: audited models/worker remain resident; other configurations reload each call. Load NR means no recorded interval, not zero. Call spans inference entry to synchronized return, including loading and native validation, excluding generic output QA. Warm peak $M_w$: maximum NVML peak over measured calls, sampled approximately every 100 ms across preparation, generation and output processing. $N$: output frames; dimensions exclude comparison panels but may exceed synthesis crops. Workloads differ. Full timing definitions and records: Section 8.7.

Twenty configurations cover 19 methods; FlashHead Lite and Pro are separate configurations. Wan's three encoded frame counts are retained in order; $528 \times 944$ neural output is cropped to $528 \times 935$. Its GPU worker persists across calls. Gen generally excludes VAE decoding; exact operators differ. Additional-source configurations and their missing settings are identified in Table 25. Moore-AnimateAnyone is third-party.

ruption scheduling [83]. Evaluation must distinguish intended changes from drift and measure transition delay, stale-reference influence, and subsequent consistency.

Conditioning imposes a latency floor before visual generation begins. Speech-encoder receptive fields and input buffering, illustrated by Teller and StreamAvatar, remain even after denoiser acceleration [176,254]. Streaming protocols should therefore pair component timing with input-chunk duration, TTFF, inter-chunk latency and jitter, playback buffering, and device count.

### 4.7 Stable: Streaming State Management

The mechanisms in Section 2.8 address different state failures. Self-Forcing++ trains on five-second segments of long self-generated rollouts and evaluates a 100 s quantitative horizon [42]. LongLive combines a short window, persistent sink, KV recaching, and streaming long tuning [218]. Prompt switches trigger recaching from recent video and the new prompt to remove stale semantics. Its 1.3B model reports 20.7 FPS at 832×480 on one H100, with 30 s single-prompt and 60 s prompt-switch tests and a separate 240 s demonstration. For virtual humans, these designs motivate recent motion state, identity anchors, compressed older history, semantic refresh, and position re-indexing. Finite-memory execution still requires output-based stability evaluation [176,218,255].

Anchors supply identity and scene priors; recent windows preserve mouth and motion continuity; older memory retrieves remote poses or events. Growing caches increase cost and retain errors, while bounded caches may forget useful history. Evaluation should track quality, legitimate changes blocked by anchors, failures, and recovery across duration bins.

InteractiveAvatar promotes semantically diverse entries from a recent FIFO buffer into a bounded long-term bank [173]. TaoMate separates active context from fixed-capacity audiovisual memory and checks recurrent appearance updates against anchors [56]. LiveAnimate combines a permanent sink, pose-matched retrieval, a rolling window, and a separate clean-KV update [244].

These policies manage different retrieval and contamination risks; ablations and duration-specific measurements test their usefulness.

Live Avatar combines an Adaptive Attention Sink (L1), Rolling RoPE (L3), and step-matched noisy history; a bounded recent window and a persistent sink retain context [81]. Its looped-speech test reports little degradation in sampled quality, synchronization, and identity windows extending beyond an hour. These observations support that repeated-audio setting; unsampled intervals, the static-scene prior, and the absence of a matched external baseline limit generalization to changing interactive sessions.

Gloria anchors scene, viewpoint, and expression across pose changes [222]. EverAnimate combines latent propagation with restorative flow matching to address texture degradation and identity drift [111]. Their tasks and durations differ (Table 9). TT-SAC instead re-encodes early outputs and generates a second pass with revised conditioning, without parameter optimization [248]. Its short-clip results and additional-pass cost remain distinct from continuous-session recovery.

Interaction capability alone does not establish stability. LPM 1.0 combines a generator, refiner, and pre-RoPE KV for speaking, listening, idling, and reflecting, but its extended demonstrations lack duration-binned quantitative evaluation [233]. FlowAct-R1 combines structured memory with generated-history replacement for planned full-body behavior [195]. In the render route, ARIG reports benefits from a larger interaction-summary cache, but capacity does not determine an evaluated generation duration [64].

### 4.8 Design Guidance

Table 10 highlights three choices. Optimize the measured bottleneck as NFE falls; use pipeline overlap for throughput and smaller initial chunks or buffers for TTFF; and evaluate cache, overlap, and compression policies jointly for their cost and retained history. No universal NFE threshold determines when decoding dominates.

### 4.9 Synthesis and Open Questions

Future studies should report the quality–speed–horizon trade-off through strict NFE, component times, TTFF, chunk latency/jitter, and memory. Cache results also require warm-up, refresh policy, and growth with duration.

A matched-compute comparison of the five stability mechanisms should measure drift, motion diversity, intended scene changes, and failures over time. TaoMate provides one reliability-aware update policy, but general recovery under unexpected inputs remains open [56]. Training comparisons should control ground-truth, generated, and mixed histories. Section 9 addresses identity-state authorization and provenance.

Diffusion systems combine compression, call reduction, reuse, and serving optimization with history management. Their joint effect is task-dependent. When a native AR backbone instead predicts visual state and diffusion is only a local emission head, the system belongs to the route discussed next.

## 5 Autoregressive Methods

The autoregressive (AR) route comprises models that own cross-frame dense or discrete visual state, as in MIDAS and Body of Her [5, 27]. AR speech, language, pose, and motion modules instead provide control when another component owns appearance. Likewise, a sequential scheduler does not make dense diffusion autoregressive. This distinction assigns the route by visible-content generation rather than system organization.

The literature nevertheless advances along two largely independent axes. The first is *state ownership*: causal state progresses from language and behavior, through motion and sparse control, to decodable visual representations. The second is *interaction scope*: one-way animation expands toward online dyadic response, full-duplex interaction, and persistent world grounding. A world-model label belongs to the second axis unless the causal model also predicts the evolving visual representation. This distinction permits highly interactive systems to inform AR system design without relabeling their flow-, diffusion-, or renderer-owned visual generation paths.

Autoregressive visual state connects efficiency directly to history management. Reusing context can avoid repeated computation, while a long context increases memory and may retain generation errors. KV caching, quantization, parallel decoding, and speculative execution provide candidate acceleration mechanisms [76, 102]. Their benefits for a virtual-human pipeline depend on the visual representation, local decoder, and quality constraints, which must be evaluated alongside temporal drift.

### 5.1 Scope, State Ownership, and Cost Boundary

The ordered test in Section 2.3 first excludes methods whose appearance is owned by a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp.

**Table 9** State management and reported generation durations in diffusion systems. Quantitative tests and demonstrations are distinguished in the final column; their duration and input conditions differ across papers.

| Method | State management | Reported evaluation and duration |
|---|---|---|
| Hallo2 [41] | Patch-drop/noisy history (L4-C) | Quantitative: 4 min, $512^2$, enhancer off; hour-long claim separate |
| StableAvatar [184] | Reference anchor; overlapping windows | Aggregate metrics on Long100: 100 clips, 2–5 min; offline generation |
| PersonaLive [113] | Selected history; capacity/eviction NR | Quantitative on LV100: at least 1 min; 2700-frame demonstration separate |
| MotionStream [166] | Sink (L1), rolling KV, L3, L4-G | Quantitative: 241 frames (∼15 s); general-motion task |
| JoyStreamer-Flash [105] | Boundary transfer (L5), L3 | Demonstration: 120 s; no time-binned quantitative test |
| StreamAvatar [176] | Sink (L1), L3, L4-G | Quantitative: 20 s; ∼150-s demonstration |
| LiveTalk [29] | Three sinks, two recent slots, L4-G | 100-s demonstration; several-minute claim |
| EchoTorrent [135] | Reference sink; tail-only forcing | 1000-s demonstration |
| Live Avatar [81] | Sink (L1), L3, L4-G; noisy KV | Sampled quantitative windows up to 10000–10010 s; repeated audio, static-scene prior |
| LPM 1.0 [233] | Sink/reference; pre-RoPE KV | 60,000-s demonstration timeline; time-binned metrics NR |
| AsymTalker [125] | Asymmetric GT/generated history | Time-specific FID/ArcFace at 5/50/150/300/600 s |
| Lip Forcing [31] | Sink, dynamic positions, L4-G | Quantitative: up to 6 min; speech pre-encoded |
| SoulX-FlashTalk [165] | Retrospective self-correction (L4-G) | Quantitative: over 5 min; 1000-s demonstration |
| SoulX-FlashHead [232] | Reference anchor; L4-G | 60-s demonstration |
| SoulX-LiveAct [255] | Fixed reference/long/recent slots (L2) | Hour-scale claim; duration-binned metrics NR |
| Avatar V [114] | Appearance anchor; motion transfer (L5) | Unbounded-length claim; quantitative duration NR |
| FlowAct-R1 [195] | Structured memory; generated-history replacement | Longest quantitative rollout NR; full-body scope |
| Gloria [222] | Scene/view/expression content anchors | Aggregate consistency on 20 clips of 5–10 min; not a time-binned drift curve |
| EverAnimate [111] | Latent propagation; restorative flow | Quantitative target durations of 10/30/60/90 s; pose-driven animation |
| Avatar-Forever [108] | Recovery rollout training; within-chunk cache | Quantitative: 5 and 30 s; cache reset for each chunk |
| LiveAnimate [244] | Fixed sink; bounded pose retrieval and rolling KV | Three-minute evaluation; reported block timing excludes VAE decoding |

Among the remaining systems, the AR route is selected only when the evolving visual tokens or latents are predicted natively from prior visual state; a downstream diffusion process may refine the current representation without changing that assignment.

For one emitted frame or block, a useful route-specific accounting is

$$C_t = C_{\text{condition}} + C_{\text{AR}}(h_t) + C_{\text{local-emission}} + C_{\text{decode/render}} + C_{\text{system}}, \tag{5}$$

where $h_t$ names the persistent visual state available at step $t$. The AR term includes token prediction, attention, and state update; a local continuous-output head retains its own denoising or flow evaluations; and the final terms include codec decoding, rendering, transfer, compositing, and input/output (I/O). Consequently, one AR-backbone update per frame does not imply only one network evaluation or one complete system call per visible frame, and a reported frame rate is interpretable only with the hardware, resolution, precision, and included stages defined in Section 2.7.

## 5.2 Mechanism Map: Compression, State, and Emission

AR visual generation exposes three coupled design choices [5, 27, 209]. First, the representation fixes serial granularity. A frame may contain many discrete tokens [139], a compact set of continuous tokens [24], or one latent block. Spatial compression and within-frame parallel prediction reduce different work, but tokenizer and decoder costs remain. Second, persistent state controls repeated computation and resource growth. KV reuse avoids recomputing past projections; windows, summaries, eviction, and resets can bound the retained state. A codec feature cache is not interchangeable with an AR-backbone context. Third, local emission converts predicted state into an observable frame. A diffusion or flow head may improve continuous output, but its calls and final decoder remain on the critical path [5, 27].

Compression shortens the AR sequence but can shift cost or error into the codec. Longer caches retain context and erroneous history; truncation bounds memory but can discard identity or conversational events. Within-frame parallelism reduces serial depth while

**Table 10** Diffusion-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

| Target | Design options | Trade-offs to check |
|---|---|---|
| Faster fixed clips | step caches, CFG folding, compressed latents | Cache staleness near detail steps; compression harms mouths and teeth |
| Playback-rate streaming | few-step DMD, rolling KV, DiT/VAE overlap, FP8 and fused kernels | Multi-GPU budgets; TinyVAE quality loss; L4-G training cost |
| Low TTFF | smaller initial chunks and staged startup | Input buffering floor; startup-quality gap |
| Minute-scale stability | anchors, generated-history training, position management | Static-scene bias; motion-diversity loss; memory policy requires validation |
| Hour-scale operation | bounded memory, anchors, positions; Potential: adaptive refresh/recovery | Check output between sampled windows and under varied input |

cross-frame self-conditioning remains vulnerable to accumulated error. Each trade-off requires separate efficiency and stability measurements.

## 5.3 Autoregression at Different Pipeline Levels

The predicted variable determines the role of an AR module. UniTalker models responses and compact facial behavior; Teller, ARIG, DyStream, Talker-T2AV, and UniLS predict motion or audiovisual controls that a renderer maps to visible appearance [22,36,64,75,225,254]. These systems motivate causal control, with rendering retained as a separate cost. OmniResponse provides a detailed example of how autoregression coordinates conversational behavior across modalities.

Native visual AR instead predicts dense visual latents or appearance-bearing tokens. UniMo interleaves 2D video and quantized 3D motion tokens, providing a general human-video bridge; MIDAS, Body of Her, and EARTalking target conditioned human video with framewise visual state [5, 27, 141, 204]. The inference-graph test and worked cases in Figure 6 replace a second taxonomy here.

### *5.3.1 OmniResponse: Autoregressive Conversational Responses*

OmniResponse generates conversational text and facial behavior from speaker/listener histories [126]. Chrono-Text Markup encodes silence and word duration to align text with facial time steps, and TempoVoice predicts audio tokens from that aligned stream. A decoder recovers facial and pose controls for a pretrained portrait renderer. The recurrent state therefore owns behavior, while the renderer owns appearance.

Removing temporal markup or replacing TempoVoice worsens lip–speech alignment under the source protocol [126]. These ablations support synchronized modality generation, but do not evaluate identity drift during long sessions.

OmniResponse's runtime report leaves timing endpoints and inclusion of rendering and I/O unspecified [126]. Its latency cannot be treated as TTFF; a complete visible-output boundary remains necessary for comparison.

## 5.4 Representative Visual-State Autoregression

UniMo, MIDAS, Body of Her, and EARTalking satisfy the visual-state criterion under different task and decoding assumptions. UniMo supplies a general human video–motion example, whereas the other systems target digital humans or talking heads. Table 11 summarizes their state representations, reported runtime, and temporal evaluation.

MIDAS predicts compressed continuous visual state before local iterative emission [24, 27]. A DC-AE limits each frame to at most 60 visual tokens, and a four-step diffusion head realizes the predicted representation before decoding. Block-level conditioning permits input look-ahead, while codec history and AR context belong to separate state owners.

MIDAS's reported frame rate and block length describe media timing, with measured throughput and inference hardware undisclosed. Its extended qualitative examples and longer-duration claim lack time-binned stability metrics [27]; Table 11 retains their distinct scopes.

Body of Her provides a framewise counterpart. Its Transformer decoder conditions on human/agent audiovisual histories, text or trajectory controls, and identity. A compact diffusion head samples the current frame's continuous tokens in parallel, followed by video decoding [5]. This separates cross-frame prediction from within-frame sampling; both contribute to visible-output cost.

**Table 11** Representative visual-autoregressive systems. Timing includes local sampling only where stated in the cited paper; NR denotes not reported. Figure 6 illustrates the generation paths.

| Method | State / local emission | Reported runtime and resources | Temporal evaluation |
|---|---|---|---|
| MIDAS [27] | Eight AR blocks; five codec frames; four-step local head | Measured FPS, TTFF, hardware NR; 25 FPS/480 ms are media rate/span | One-hour author claim; four-minute qualitative demonstration; time-binned metrics NR |
| Body of Her [5] | Frame-causal state; compact parallel diffusion head; cache bound NR | 42 ms/next frame at 640 × 360; device and included stages NR; 24 FPS media rate | Long-session quantitative duration NR |
| UniMo [141] | Visual and motion tokens; deployment cache contract NR | General human video–motion bridge; conversational service timing NA | Conversational long-session evaluation NA; general-task results separate |
| EARTalking [204] | Ten-latent-frame window; 36 spatial AR and 25 denoising steps | $512^2$ setup; numeric FPS, TTFF and inference device NR; source notes real-time limitation | Anchor/history-corruption ablations; demonstration at 3× training window; minute-scale binned metrics NR |

Body of Her's timing lacks an inference device and complete component boundary; its agents also require subject-specific data [5]. The personalized architecture leaves controlled throughput, generalization, and long-session stability comparisons open.

EARTalking makes framewise persistence explicit: a GPT-style causal backbone predicts the next 3D-VAE frame representation using prior visual context, frame-level control, a sink-frame anchor, a sliding KV window, and cyclic positions [204]. The backbone owns evolving visual state, while frame-local generation and decoding remain separate cost terms.

Runtime must follow the same ownership boundary. DyStream measures audio-to-motion delay within a render-owned system [22], whereas native visual owners must also account for visual-state prediction and emission. Table 11 separates the state representation from reported runtime and temporal evaluation.

*Measured coverage of autoregressive components.* No native visual-AR implementation is measured. DyStream and Talker-T2AV retain renderer-owned appearance and appear in Table 15. UniLS is reported separately as motion prediction plus FLAME visualization (Table 12); its measured decoding, rendering, smoothing, and export do not constitute photorealistic visual autoregression.

*Full-duplex systems as contextual interfaces.* X-Streamer and Wan-Streamer separate interaction state from dense visual generation. X-Streamer's Thinker conditions an iteratively denoising video Actor [208]. Wan-Streamer v0.2 combines a token-causal Thinker with a context-parallel flow-matching Performer; v0.3 adds persistent world context and localized events [77, 78]. These systems inform interaction design while retaining diffusion- or flow-owned visual generation.

### 5.5 Fast Design Guidance

The Fast levers below are tied to the cost terms in Equation (5). Examples from native visual AR illustrate representation and state choices; contextual motion and diffusion systems illustrate related execution strategies.

- **Compress the serial visual state** ($C_{\mathrm{AR}} \downarrow$). MIDAS limits a frame to at most 60 DC-AE tokens, while UniMo interleaves compressed video and quantized motion tokens. Compression can reduce serial work but may move cost or error into the codec, particularly for fine facial detail and rapid articulation.
- **Parallelize local emission** (serial depth ↓). Body of Her samples all continuous tokens of the current frame in parallel through a compact diffusion head, and MIDAS uses a four-step local head after the AR frame state. EARTalking similarly separates cross-frame causal state from frame-local realization. The local head's strict number of function evaluations (NFE), tensor shape, and decoder cost remain part of end-to-end latency.
- **Reuse and bound persistent state** ($C_{\mathrm{AR}}(h_t)$ growth control). Standard KV reuse avoids recomputing past projections. MIDAS truncates AR context to eight blocks while separately retaining five codec frames; EARTalking combines a sink-frame anchor with a sliding KV window and cyclic positions. These mechanisms bound repeated work, but their eviction, correction, and reset costs require direct measurement.
- **Separate control-path and emission-path serving** ($C_{\mathrm{system}} \downarrow$). Wan-Streamer v0.2 keeps its token-causal Thinker on one GPU and places high-resolution flow-matching generation in a Ulysses context-parallel Performer [77]. This diffusion system places the causal state update and the expensive local emitter on distinct serving paths organized

**Table 12** Our single-H200 measurements of 1 motion-visualization configuration after three complete warm-ups. Time in seconds; sampled memory in GiB.

| Configuration | $N$ | $W \times H$ | Pre | Gen | Post | Load | Call $\pm s$ | Warm peak $M_w$ (GiB) |
|---|---|---|---|---|---|---|---|---|
| UniLS [36] | 32 | 512×512 | 0.01 | 0.15 | 0.60 | 2.09 | $2.88 \pm 0.98$ | 3.73 |

Means over three measured calls after three complete same-interpreter warm-ups; Call includes sample standard deviation. $R$: audited models/worker remain resident; other configurations reload each call. Load NR means no recorded interval, not zero. Call spans inference entry to synchronized return, including loading and native validation, excluding generic output QA. Warm peak $M_w$: maximum NVML peak over measured calls, sampled approximately every 100 ms across preparation, generation and output processing. $N$: output frames; dimensions exclude comparison panels but may exceed synthesis crops. Workloads differ. Full timing definitions and records: Section 8.7.

UniLS predicts motion in Gen and performs FLAME decoding, rendering, smoothing and export in Post. This is a motion visualization, separate from photorealistic appearance generation; native visual-AR coverage remains zero.

by a pipelined unit schedule, without changing the route assignment.

– **Test speculative visual decoding as a transfer hypothesis**. Speculative decoding is established for language models and has been studied for general AR video [76,102]. A virtual-human deployment must additionally verify identity, lip synchronization, and visual-state acceptance when drafted tokens or frames are committed; none of the retained native visual systems reports such a comparison.

Published runtime values cover different paths: DyStream reports motion-level delay, X-Streamer reports a multimodal system rate, and Wan-Streamer v0.2 reports model-side and network-inclusive interaction latency [22, 77, 208]. An AR deployment comparison should instead disclose one complete pipeline: AR calls and tokens per visible frame, local-head NFE, cache-update cost, decoder time, I/O, and peak state memory.

## 5.6 Stable Design Guidance

Visual autoregression repeatedly consumes committed visual state, which can preserve continuity and propagate error. The common patterns therefore require method-specific checks. EARTalking documents an L1 reference sink, L3 cyclic positions, and L4-C noise-corrupted histories. MIDAS also uses L4-C; its five-frame codec history and eight-block AR context are different state objects and do not alone establish multiscale L2 memory. Body of Her's dataset-history training is distinct from generated-history training. None of these facts should be relabeled as an L4-G rollout study.

Explicit boundary transfer (L5) likewise requires a designated motion or boundary variable; ordinary access to preceding visual tokens is insufficient. Generated-history training, adaptive reference refresh, and hierarchical state are research directions whose effects on long-session output remain to be measured.

**Table 13** Autoregressive design options and their trade-offs. Potential extensions require evaluation in the intended setting.

| **Target** | **Design options** | **Trade-offs to check** |
|---|---|---|
| Low serial latency | compression and parallel within-frame emission | Codec error on facial detail |
| High throughput | KV reuse; Potential: speculative visual decoding | Draft-model identity fidelity unknown |
| Minute-scale stability | anchors and bounded history; Potential: generated-history training and adaptive refresh | Accumulated error; anchor over-constraint; refresh discontinuity; matched long-session comparison still needed |
| Hour-scale operation | Potential: hierarchical state and declared eviction/reset/recovery | Long-range forgetting; reset discontinuity; unmeasured memory and error trajectories |

EARTalking's component ablations provide a concrete example, analyzed in Section 7; the retained systems do not yet provide a complete joint time-binned session protocol.

## 5.7 Synthesis and Open Questions

Visual AR systems differ in representation, personalization, and local emission. None in Table 11 combines complete end-to-end timing with a joint time-binned stability evaluation. The evidence therefore establishes architectural feasibility more broadly than deployment comparability.

Co-design should connect tokenizer compression to decoder cost, expose state growth and reset policies, and count local-head NFE within the complete event timeline. Sub-linear-state sequence models [59,221] offer

constant-size state but require evaluation of forgetting and correction. Response- and motion-level autoregression remain useful interfaces whose appearance costs belong to their visual owners.

The route's central choices are visual compression, history reuse, and local emission. Matched timing and stability measurements are needed to determine how each choice changes computation and error propagation.

## 6 Render-Based Methods

### 6.1 Scope, Ownership, and Cost Boundary

Render-based methods animate reusable appearance through time-varying controls. A subject-specific mesh, neural field, Gaussian set, or identity-conditioned image-space representation owns appearance, and its renderer produces visible frames [62, 63, 91]. The controller may be deterministic, autoregressive, diffusion-based, or flow-based; its mechanism does not change the appearance owner's route.

Reusing appearance connects the route's efficiency and stability mechanisms. An enrolled asset reduces repeated appearance construction and supplies a persistent identity reference, corresponding to L1 in Section 2.8. Frame production still requires deformation, rendering, and often neural decoding or inpainting. Enrollment, tracking, and compositing must therefore be included when assessing the complete pipeline. For a session of $T$ observable frames, we write

$$\begin{aligned} C_{\text{session}} = C_{\text{enroll}} + \sum_{t=1}^{T} \big[ & C_{\text{track/cond}}(t) + C_{\text{motion}}(t) \\ & + C_{\text{repr}}(t) + C_{\text{render}}(t) \\ & + C_{\text{compose}}(t) + C_{\text{system}}(t) \big]. \end{aligned} \tag{6}$$

Here, $C_{\text{enroll}}$ covers capture, calibration, tracking, reconstruction, rigging, per-subject optimization, and asset storage; $C_{\text{track/cond}}$ covers runtime acquisition and estimation of pose, camera, gaze, audio, or other controls; and $C_{\text{motion}}$ is the cost of predicting the compact time-varying state. The remaining terms cover appearance deformation or attribute updates, rendering, masks/inpainting/super-resolution, and system overhead such as transfer, scheduling, encoding, and I/O. Thus, renderer-only throughput cannot be reported as end-to-end latency, and a high per-frame rate cannot hide a costly enrollment stage.

Figure 9 projects Equation (6) onto four recurring interface choices. The representations expose different computational boundaries: enrollment and reused appearance are session-level, whereas the control-to-output path repeats per frame, and individual methods may omit or relocate the optional source-pixel and compositor paths.

Codec avatars provide an early example of person-specific learned appearance for real-time telepresence under studio capture conditions, including deep appearance models [121], expressive extensions [33], and audio- and gaze-driven animation [157]. Subsequent work explores different enrollment budgets, from multi-view capture to monocular video and single-image reconstruction. These settings offer different trade-offs between acquisition effort, subject fidelity, and rendering cost.

### 6.2 Explicit Controllable Appearance Representations

Explicit backends update geometry or scene attributes from pose, expression, viewpoint, or audio controls. 3DMM, FLAME, SMPL, and SMPL-X supply interpretable interfaces [9, 109, 123, 145]; RC-SMPL progressively accumulates texture and normal maps from a single RGBD camera [170]. NeRF and 3DGS provide appearance representations [91, 137], while capture coverage, deformation, compositing, and hardware determine each avatar's fidelity and runtime.

#### *6.2.1 Neural-field backends*

Early neural fields made appearance target-specific. AD-NeRF learns separate audio-conditioned head and torso fields from a monocular target video [63]. Head-NeRF renders a low-resolution feature map followed by neural upsampling [71], while FLAME-in-NeRF adds parametric expression control [6]. Their reported rendering rates exclude different fitting and system costs; Table 14 retains those boundaries.

Hash encodings and occupancy pruning reduce field-query work while retaining subject enrollment. Following Instant-NGP [138], RAD-NeRF combines spatial and audio grids with a separate torso stage [182]; ER-NeRF uses tri-plane hashes and regional attention [107]. Their runtime reports use different devices, enrollment procedures, and timing boundaries, so they do not isolate a common speedup over AD-NeRF.

SyncTalk improves the conditioning path around a personalized tri-plane field: Face-Sync aligns articulation, Head-Sync stabilizes pose, and Portrait-Sync restores detail and blends the torso [148]. These synchronization and composition modules affect the completed video beyond field-rendering cost.

AvatarReX extends personalized fields to separately controlled body, hands, and face through surface-based

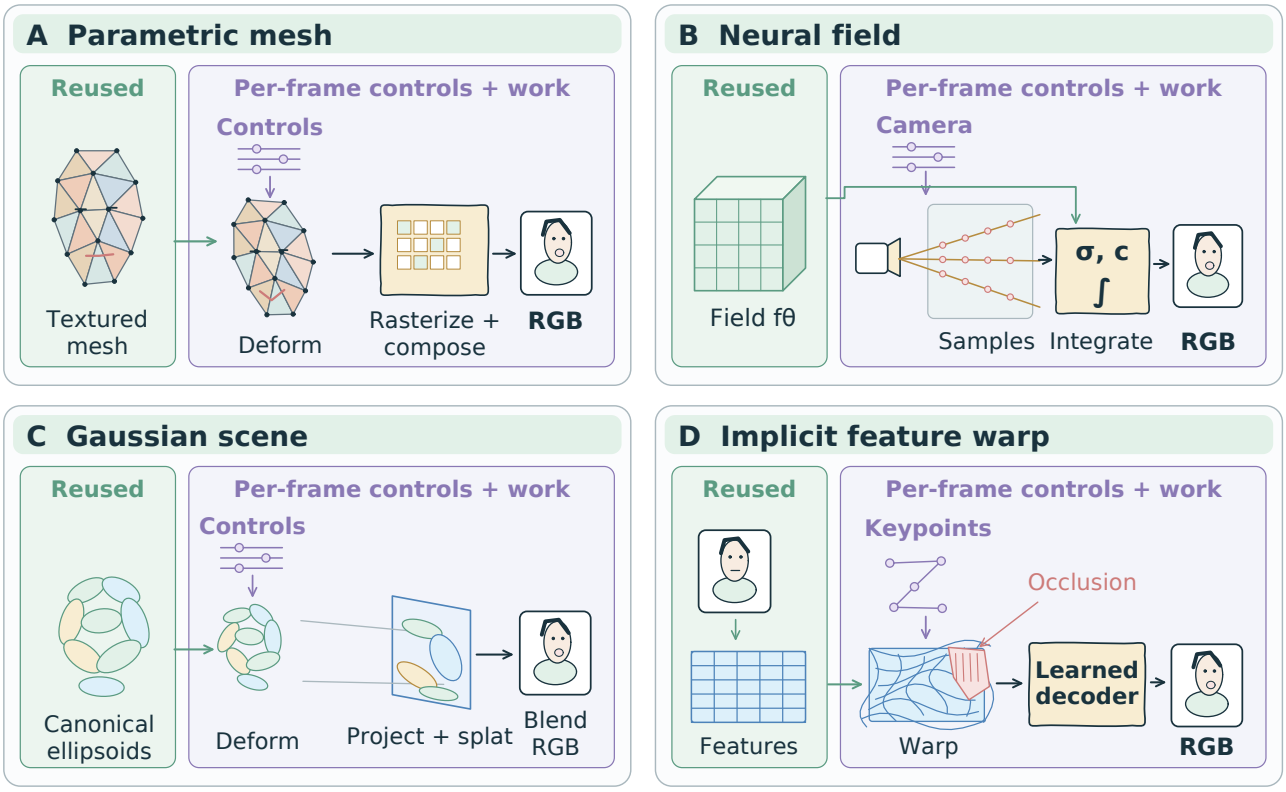



**Fig. 9** Reusable appearance in four representations: (A) a textured parametric mesh, (B) a neural field sampled along rays, (C) Gaussians projected and rasterized, and (D) a source feature volume transported by a learned warp [62,63,107,109,168,206,229,249]. Each panel distinguishes the stored appearance from the control and rendering operations repeated for each frame. The diagrams explain representations rather than rank speed, quality, or stability.

deferred rendering [256]. Its studio multi-view enrollment illustrates the acquisition cost behind efficient full-body rendering.

### *6.2.2 Gaussian scene representations*

Gaussian avatars replace repeated field sampling with rasterizable primitives. Yu et al.'s GaussianTalker binds a target face to FLAME, predicts speaker-specific blendshapes, and composites the rendered face with preserved source pixels [229]. GaussianAvatars attaches multi-view Gaussians to FLAME triangles [152]; SplattingAvatar combines mesh deformation with walkable barycentric Gaussians [163]; and FlashAvatar uses compact surface-embedded Gaussians with residual deformation [206]. Related designs vary blendshape latents [50], shape models [253], body deformation [153], emotional control [217], progressive construction [171], full-pipeline serving [99], toonification [172], and hand animation [250]. Efficient rasterization leaves enrollment, tracking, and topology handling as separate costs.

Audio-driven splatting can also bypass explicit blendshape control. Cho et al.'s GaussianTalker predicts canonical-Gaussian offsets through spatial–audio attention [30], distinct from Yu et al.'s FLAME-bound method with the same name [229]. SyncTalk++ moves the synchronization-oriented design to dynamic Gaussians and adds out-of-distribution audio expression generation and torso restoration [147]. Its changed hardware and auxiliary modules prevent attributing the reported throughput gain solely to the renderer.

Recent controllers animate richer Gaussian state. GaussianSpeech predicts expression-dependent attributes; DGTalker separates lip motion from broader expression; and GGTalker combines a generalizable prior with identity adaptation [4,74,115]. AudioAvatar extends audio-conditioned deformation to face, hands, and body, using diffusion-feature distillation during training [101]. Gaussian appearance remains the visual owner, while out-of-distribution speech, large motion, and deformation introduce additional stability risks.

The deformation representation itself can reduce recurring computation. EmbedTalk replaces a triplane encoder with per-Gaussian learned embeddings that

are deformed using audio and facial controls [159]. This removes a feature-lookup stage while retaining per-identity optimization of the Gaussian appearance model. Its enrollment requirement therefore differs from one-shot reconstruction, even though both approaches can use the same rasterization primitive.

GaussianHeadTalk studies a complementary temporal failure mode: inconsistent Gaussian mapping and noisy facial tracking can produce wobble despite a fixed identity asset [1]. This motivates separate measurements of tracking and deformation error, failed detections, and temporal artifacts, alongside identity and synchronization.

Gaussian systems now address both wearable execution and one-shot construction. TaoAvatar bakes deformation into a lightweight network for augmented-reality glasses [25]; SqueezeMe removes recurring corrective computation to support multiple avatars on a headset [84]. Sustained power and thermal behavior remain separate evaluation targets. LAM, GUAVA, and VASA-3D instead amortize appearance construction into feed-forward networks [67, 211, 235]. Their reduced input requirements leave unseen appearance and subject-specific detail dependent on learned priors.

SDTalk extends one-shot Gaussian construction by separating visible-surface reconstruction from prior-based completion of occluded facial regions, then using coarse and fine motion fields [87]. It requires no personalized fine-tuning, but the teeth and other unseen structures still depend on the learned prior. Lower enrollment cost should therefore be assessed alongside the fidelity of newly exposed regions.

GPAvatar reconstructs a controllable tri-plane field from one or several images, followed by volume rendering and super-resolution [35]. GAGAvatar keeps generalizable FLAME control but reconstructs Gaussians through dual-lifting [34]. Their reported timing difference changes reconstruction and rendering together, so it is not an isolated representation ablation. GAGAvatar also provides an appearance backend for downstream motion controllers.

### 6.3 Implicit Image-Space Appearance Reuse

Image-space methods reuse a source image or feature volume through learned correspondences. FOMM predicts keypoints, local Jacobians, dense flow, and occlusion for warping and inpainting [168], extending keypoint transfer and articulated-region modeling [167, 169]. TPS Motion Model replaces local affine motion with thin-plate-spline transformations [249]. Face-vid2vid uses 3D-aware keypoints for one-shot animation and low-bandwidth transmission [199]; latent navigation [200] and audio-driven control [197] provide alternative motion interfaces. All retain source-dependent appearance and must synthesize disoccluded regions.

LivePortrait warps a source feature volume with learned implicit motion and decodes it with a SPADE-based generator, supporting cross-identity animation and paste-back [62]. Its model-only timing omits a complete preparation and composition breakdown. Large pose changes, mouth interiors, hair, disocclusion, and crop boundaries remain relevant failure cases in systems that reuse this backend.

Reference reuse can also target oral detail. EfficientSync selects source-video references, warps their features, mixes their textures, and decodes the mouth region with a SPADE generator [70]. The transported reference appearance places it in the render route despite adversarial training. Reference selection belongs to preparation cost, and missing views or mouth states can limit the texture information available at inference.

### 6.4 Motion Controllers over Reused Appearance

Reusable appearance permits motion control to be studied separately from image synthesis. A compact motion model can drive a one-shot warp or Gaussian asset, changing temporal behavior without changing the visual owner. The controller and renderer have separate costs, and either can determine system latency.

VASA-1 samples holistic facial dynamics in a learned motion space and drives a one-shot appearance decoder [210]. FLOAT and FlowTalk use motion-space flow matching [47, 95]; Ditto structures controllable motion [110]; ChatAnyone and MirrorMe extend the interface to upper- or half-body animation [133, 151]. MoDA couples rectified-flow motion generation to LivePortrait's frozen appearance modules [112]. GAIA scales the motion–appearance decomposition [66], JoyVASA extends it to animal faces [18], and style-conditioned variants enrich motion control [130, 198]. Speech-to-3D systems remain motion-only boundary components when they predict vertices or coefficients without producing appearance [54, 251]. MambaTalk combines selective state-space modeling with discrete motion priors and local/global scans for holistic gesture synthesis, providing another compact motion interface [213].

Compact controllers now separate articulation from other aspects of personal motion. VibeAvatar generates motion in LIA's warp space, adapting phonetic conditions before preference-based optimization of motion aesthetics [196]. Talk Like You predicts habit-conditioned motion in one sampling step and uses

**Table 14** Selected rendering systems and reported operating conditions. Frame or rendering time differs from TTFF; the final column identifies which components are measured.

| Method | Appearance / enrollment | Source timing / device | Measured stages and temporal evaluation |
|---|---|---|---|
| AD-NeRF [63] | Head/torso fields; 3–5-min capture | ∼12 s/frame, $450^2$, RTX 3090 | Frame rendering; ∼36 h training |
| StyleAvatar [194] | Person-specific UV/background; adversarial decode | 35 FPS / 28 ms live; RTX 3090 | 20 ms model-only; hours-scale enrollment; longitudinal metrics NR |
| FlashAvatar [206] | FLAME-surface Gaussians; 1–3-min capture | 304 FPS, $512^2$, RTX 3090 | Animation/render only; tracking excluded |
| TaoAvatar [25] | Full-body Gaussians; multi-view capture | 90 FPS, Apple Vision Pro | Complete pipeline coverage NR |
| LivePortrait [62] | Source feature volume; one-shot | 12.8 ms model generation; RTX 4090 | Warping/decoding; preprocessing and paste-back separate |
| VASA-1 [210] | Reused face appearance; one-shot | Up to 40 FPS, $512^2$; RTX 4090 | 170/95 ms for one/two devices under source protocol |
| Teller [254] | Source image features | 7/106/71 ms components; four-H800 machine | Total $<$ 200 ms; TTFF and full RGB boundary NR |
| DyStream [22] | Reference appearance volume; one-shot | 29.24 FPS; 34.2 ms packet delay; H200 | Timer ends at motion; 180-s pose-drift ablation |
| AvatarReX [256] | Body/hand/face fields; 22-camera capture | 25 FPS, $1024^2$, RTX 3090 | Animation/render; enrollment and tracking separate |
| GAGAvatar [34] | Feed-forward Gaussian head; one-shot | 67 FPS, $512^2$, A100 | Driving/render; driving-parameter acquisition excluded |
| MoDA [112] | Motion flow + LivePortrait; one-shot | RTF 0.846; inference device NR | Wild-data protocol; training hardware is not inference hardware |

frozen LivePortrait rendering modules [187]. Both remain appearance-reuse systems: richer motion objectives change the controller without turning it into a dense visual generator. Evaluation should distinguish synchronization, motion preference, and reproduction of a target speaking habit.

Interactive agents add session state to reusable appearance. Teller couples autoregressive quantized motion to keypoint warping and enhancement [254]; ARIG combines framewise control, a local diffusion head, and bounded interaction summaries [64]; DyStream drives a reference-conditioned appearance volume with AR/flow motion [22]. REA-Listener uses a lightweight controller with GAGAvatar/3DGS and inpainting [34,252]. PortraitDirector adds disentangled reenactment controls [85], and VASA-Rig connects audio-driven rigging to virtual reality [140].

EmbodiedHead's reported full-pipeline rate is substantially lower than its motion-only rate [243]. This within-source comparison shows why controller timing cannot substitute for completed video generation.

## 6.5 Open-Source Measurements on One H200

Table 15 reports repeated measurements of 24 configurations. Generation includes both controller and renderer where present. Input lengths and supplied assets differ; released subject training and precomputed tracking are excluded when the demo accepts them as prerequisites (Appendix A). Only StyleAvatar retains audited neural objects in this group; the other 23 configurations reload models each call.

The repeated measurements expose work beyond rendering: LAM averages 14.07 s in fresh tracking/preparation and 0.97 s in Gen, while GUAVA averages 50.92 s in output processing versus 9.45 s in Gen. SDTalk similarly spends 33.92 s in Post and 12.42 s in Gen. These decompositions connect Equation (6) to the implementation path without ranking the representations.

Some configurations need distinct interpretation. StyleAvatar retains its style generator and stateless ORT sessions but rebuilds FaceVerse tracking/fitting state each call; its two-frame output cannot estimate sustained throughput. SplattingAvatar uses the released Python evaluation path rather than its optimized Unity runtime, and RAD-NeRF uses the repository's PyTorch reimplementation. Ditto and Talk Like You have overlapping stages, while SyncTalk performs native extra validation inside the call window. Their phase unions and additional costs are retained separately.

## 6.6 Fast Design Guidance

The route's levers map onto Equation (6) as follows:

1. **Amortize appearance at enrollment** ($C_{\text{repr}}/C_{\text{render}} \downarrow$ via $C_{\text{enroll}} \uparrow$). Subject-specific fields and Gaussian sets reuse fitted appearance

**Table 15** Our single-H200 measurements of 24 render-route configurations after three complete warm-ups. Time in seconds; sampled memory in GiB.

| Configuration | $N$ | $W \times H$ | Pre | Gen | Post | Load | Call $\pm s$ | Warm peak $M_w$ (GiB) |
|---|---|---|---|---|---|---|---|---|
| StyleAvatar$^R$ [194] | 2 | 1024×1024 | 7.74 | 0.04 | 0.46 | NR | 8.25 ± 0.12 | 5.94 |
| AD-NeRF [63] | 16 | 450×450 | 0.12 | 93.43 | 0.21 | 0.06 | 93.88 ± 1.60 | 2.12 |
| RAD-NeRF [182] | 588 | 450×450 | 1.47 | 4.90 | 2.81 | 0.04 | 9.24 ± 0.04 | 1.24 |
| ER-NeRF [107] | 16 | 450×450 | 0.11 | 0.19 | 0.42 | 0.95 | 1.69 ± 0.01 | 1.44 |
| SyncTalk [148] | 16 | 512×512 | 7.36 | 0.22 | 0.35 | 0.47 | 14.05 ± 0.12 | 1.74 |
| GPAvatar [35] | 16 | 512×512 | 1.12 | 0.77 | 0.92 | 1.31 | 4.14 ± 0.04 | 11.42 |
| LAM [67] | 51 | 512×512 | 14.07 | 0.97 | 0.96 | 15.73 | 31.95 ± 2.53 | 5.82 |
| FOMM [168] | 169 | 256×256 | 3.19 | 1.19 | 0.46 | 0.66 | 5.52 ± 0.32 | 2.15 |
| TPS Motion Model [249] | 169 | 256×256 | 3.45 | 1.45 | 0.45 | 0.66 | 6.02 ± 0.64 | 2.10 |
| LivePortrait [62] | 78 | 600×704 | 2.15 | 1.26 | 2.34 | 2.33 | 8.20 ± 0.39 | 4.16 |
| LIA [200] | 652 | 256×256 | 1.08 | 12.66 | 5.76 | 1.22 | 20.74 ± 1.59 | 4.71 |
| FLOAT [95] | 376 | 512×512 | 0.03 | 4.14 | 6.17 | 5.69 | 16.04 ± 0.27 | 8.40 |
| Ditto [110] | 394 | 1432×1432 | 0.70 | 18.17 | 13.96 | 10.72 | 33.43 ± 2.44 | 9.42 |
| MoDA [112] | 375 | 624×624 | 0.37 | 7.29 | 6.89 | 10.85 | 25.43 ± 3.27 | 48.55 |
| JoyVASA [18] | 321 | 512×512 | 0.63 | 7.54 | 6.12 | 3.99 | 18.42 ± 2.01 | 7.12 |
| DyStream [22] | 235 | 512×512 | 0.66 | 16.38 | 3.70 | 10.19 | 30.99 ± 1.93 | 19.52 |
| Talker-T2AV [225] | 136 | 512×512 | 0.09 | 29.00 | 0.72 | 24.13 | 54.16 ± 0.96 | 20.48 |
| Talk Like You [187] | 495 | 1034×1034 | 4.46 | 12.52 | 10.92 | 16.42 | 35.24 ± 2.47 | 12.65 |
| SplattingAvatar [163] | 16 | 1080×1080 | 0.27 | 1.10 | 0.58 | 2.72 | 4.68 ± 0.24 | 1.65 |
| 3DGS-Avatar [153] | 16 | 512×512 | 0.99 | 0.40 | 0.99 | 2.85 | 5.70 ± 0.88 | 0.71 |
| GAGAvatar [34] | 241 | 512×512 | 2.87 | 3.22 | 12.71 | 8.25 | 27.18 ± 1.09 | 2.29 |
| GUAVA [235] | 425 | 512×512 | 12.47 | 9.45 | 50.92 | 23.58 | 96.49 ± 4.51 | 6.03 |
| SDTalk [87] | 744 | 512×512 | 5.51 | 12.42 | 33.92 | 8.28 | 60.22 ± 2.15 | 2.67 |
| face-vid2vid-unofficial [199] | 169 | 256×256 | 4.73 | 6.44 | 0.61 | 2.34 | 14.14 ± 1.14 | 4.24 |

Means over three measured calls after three complete same-interpreter warm-ups; Call includes sample standard deviation. $R$: audited models/worker remain resident; other configurations reload each call. Load NR means no recorded interval, not zero. Call spans inference entry to synchronized return, including loading and native validation, excluding generic output QA. Warm peak $M_w$: maximum NVML peak over measured calls, sampled approximately every 100 ms across preparation, generation and output processing. $N$: output frames; dimensions exclude comparison panels but may exceed synthesis crops. Workloads differ. Full timing definitions and records: Section 8.7.

Gen includes controller and renderer where present. Ditto and Talk Like You overlap stages; their unions, rather than column sums, measure the inference-stage wall interval. SyncTalk's native validation is in Call but outside Pre/Gen/Post. StyleAvatar has only two frames and rebuilds FaceVerse state each call; its style generator and ORT sessions remain resident. Supplied assets and internal warm-ups are listed in Table 26.

across frames. One-shot warps and feed-forward Gaussian builders, such as LivePortrait and LAM, reduce preparation cost but rely more on learned priors for unseen content.

2. **Structure the queries** ($C_{\text{render}}\downarrow$). Hash grids and occupancy pruning reduce field-query work, while Gaussian representations permit rasterization. Mobile and wearable examples include SplattingAvatar, TaoAvatar, and SqueezeMe. Their reported frame rates establish particular device configurations, not a measured speed-per-watt ranking.
3. **Keep the controller compact** ($C_{\text{motion}}\downarrow$). Motion-space diffusion or flow reduces the representation processed by an iterative sampler. The renderer and compositor remain on the inference path and require their own component measurements.
4. **Reuse source pixels** ($C_{\text{compose}}$ trade). Paste-back and outside-mask retention (GaussianTalker, LivePortrait) avoid re-synthesizing static content, at the price of seams and crop-boundary failures.

Table 14 shows why these levers must be evaluated separately: reported quantities span renderer-only throughput, animation-plus-render time, model-generation time, motion-level packet delay, and block latency, and no cross-paper rank is supported when anatomy, personalization, device, resolution, and compositing differ. The route-specific accounting rule is to include $C_{\text{track/cond}}$ wherever runtime face tracking, FLAME fitting, keypoint estimation, or audio conditioning is required. Precomputed controls and runtime-estimated controls define different boundaries.

### 6.7 Stable Design Guidance

An enrolled asset anchors identity (L1), while tracking, control, deformation, disocclusion, and compositing can still introduce temporal errors. Controller memories, including ARIG's 512-slot cache, retain a separate state. Their position policies and training histories require longitudinal evaluation; renderer throughput and asset reuse alone do not establish stability.

DyStream's anchor ablation measures head-pose drift over ten 180-s clips [22] (Table 14). It supports anchor conditioning for pose drift, while leaving joint, duration-binned identity, appearance, synchronization, geometry, flicker, memory, and latency untested.

A useful comparison would evaluate a tracked-and-rendered avatar and an anchored dense generator under the same long-session conditions. Identity retention, motion and deformation error, temporal wobble, and compositing failures should be reported separately. GaussianHeadTalk motivates the distinction between identity similarity and temporal wobble [1]; the reviewed studies do not establish which complete pipeline is more stable over an hour.

**Table 16** Rendering-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

| Target | Design options | Trade-offs to check |
|---|---|---|
| Per-subject reuse | Gaussian assets and baked deformation | Hours of enrollment; capture requirements |
| One-shot generality | LivePortrait warps, LAM Gaussians, VASA-1 motion controller | Disocclusion, large poses, mouth interiors; limited input-view coverage |
| Mobile/ wearable deployment | splatting and baked correctives (TaoAvatar, SqueezeMe) | Per-subject assets; thermal/battery limits untimed |
| Minute-scale stability | enrolled L1 asset; Potential: joint long-session reliability | Temporal errors in deformation and composition; limited joint long-session evaluation |
| Expressive control | motion-space diffusion/flow controller | Controller expressiveness bounded by the asset's deformation space |

### 6.8 Synthesis and Open Questions

Subject-specific fields and Gaussians amortize enrollment over many frames. One-shot warpers reduce enrollment but depend on source-view coverage and learned disocclusion. Parametric controls offer editable interfaces within a limited motion space. Explicit scenes must manage geometry, memory, and coverage; image-space reuse must reconstruct newly visible content and preserve crop seams.

Three directions follow: jointly optimize appearance construction and compact motion control; test recurrent controllers under generated histories; and compare complete pipelines over sustained sessions. Reporting enrollment data and time, asset size, hardware, precision, runtime stages, and quality would reveal whether an optimization reduces total cost or transfers it elsewhere.

Appearance reuse supports desktop, mobile, and wearable deployments, but its identity anchor leaves tracking, deformation, and compositing errors unresolved. Table 14 therefore separates enrollment, runtime, and temporal evidence.

## 7 Cross-Route Analysis and Design Trade-offs

The four routes reduce computation and maintain temporal consistency in different ways, but they share several design trade-offs. This section examines decoder choice, pipeline parallelism, history management, and appearance reuse through representative studies. The first four cases use comparisons within the same source paper. We then relate those findings to our stage-resolved implementation measurements, which retain their own workload and software settings.

### 7.1 A Faster Decoder Can Change the Quality Trade-off

PersonaLive's decoder ablation reports higher throughput with TinyVAE but weaker identity, expression, and temporal quality than SD-VAE [113]. Table 17 retains the source values. This is a local speed–quality trade-off, not a universally preferable decoder.

The deduction is that reducing denoiser work changes the value of optimizing other stages. A decoder replacement can improve total throughput while altering the very identity and temporal properties the system is meant to preserve. This experiment does not provide a complete component-time breakdown, so it does not by itself prove that decoding dominates runtime.

The useful design rule is to profile the remaining path and measure quality at the chosen operating point.

**Table 17** Decoder ablation results reported in PersonaLive [113], using one H100 at $512^2$ resolution. ID is identity similarity; tLP is temporal LPIPS.

| Decoder | FPS ↑ | ID ↑ | tLP ↓ |
|---|---|---|---|
| SVD-VAE | 11.4 | 0.6920 | 12.97 |
| SD-VAE | 15.8 | 0.6983 | 12.83 |
| TinyVAE | 20.0 | 0.6758 | 14.66 |

## 7.2 Throughput and Startup Follow Different System Paths

Live Avatar's serving ablation separates distillation, additional GPUs, timestep pipelining, VAE parallelism, and kernel optimization [81]. Table 18 reproduces the H100 results from the paper's serving ablation. Timestep pipelining improves throughput but increases startup relative to the one-GPU distilled system; later changes improve both.

The final operating point depends on algorithm and serving topology together. Its gain over single-GPU distillation is a system-configuration comparison, while the kernel row bundles several changes. Neither isolates a same-resource algorithmic speedup or an individual kernel's contribution.

**Table 18** Serving ablation results reported in Live Avatar [81], at 720 × 400 resolution. TTFF excludes networking; NFE includes cache refresh. †: A separate table in the source reports 0.94 s for the default block size; the configurations are not fully aligned.

| Configuration | GPUs | NFE | FPS | TTFF (s) |
|---|---|---|---|---|
| Teacher | 1 | 80 | 0.29 | 45.50 |
| Distilled | 1 | 5 | 3.66 | 4.56 |
| Sequence parallel | 4 | 5 | 4.50 | 3.94 |
| Timestep pipeline | 4 | 4 | 10.16 | 4.73 |
| + VAE parallel | 5 | 4 | 20.88 | 2.89 |
| + kernel optimization | 5 | 4 | 45.20 | 1.21† |

*Enrollment changes the accounting horizon.* For additive resource cost, the amortized cost per frame is $\bar{C}(N) = C_{\text{enroll}}/N + C_{\text{runtime}}$. As an illustrative calculation on one fixed device, a reusable system with 3600 GPU-seconds of enrollment and 0.02 GPU-seconds per frame breaks even with a 0.10-GPU-second direct generator after $3600/(0.10 - 0.02) = 45{,}000$ frames. This is 30 minutes of content at 25 FPS. The numbers are hypothetical; the calculation shows why enrollment feasibility, expected reuse, and representation coverage must all enter a deployment decision.

## 7.3 An Ablation Supports a Mechanism Within Its Tested Horizon

EARTalking's ablations support corrupted-history training (L4-C) and reference anchoring (L1): removing either degrades source-reported fidelity and synchronization, with a qualitative example extending to three training-window lengths [204]. Throughput, TTFF, inference hardware, and minute-scale time-binned stability remain unreported in the paper.

The ablation shows that anchoring and history corruption improve the tested outputs. Measuring how these gains evolve during a longer session requires additional time-dependent evaluation. Similarly, Live Avatar's sampled windows under repeated audio support a specific extrapolation test, not every intervening instant or every changing-input session. Bounded state establishes a resource property, while the outputs establish the tested stability property.

## 7.4 Reusing Appearance Leaves Other Failure Paths

Rendering stores appearance separately from the motion and tracking stages that can introduce temporal errors. DyStream's anchor ablation evaluates head-pose drift across ten three-minute clips [22]; GaussianHeadTalk measures temporal wobble and illustrates mouth-detail errors under shorter reenactment protocols [1]. These studies examine different temporal failures. Identity, synchronization, and sustained service reliability require additional measurements.

The architectural deduction is that a persistent appearance representation narrows what must be regenerated but leaves the controller, tracker, deformation field, and compositor responsible for temporal behavior. The corresponding research hypothesis is that generated-history training and error-triggered refresh could improve sequential controllers without changing the appearance owner. This proposed transfer requires controlled testing. Figure 10 grounds the failure vocabulary in source-reported examples rather than schematic degradation.

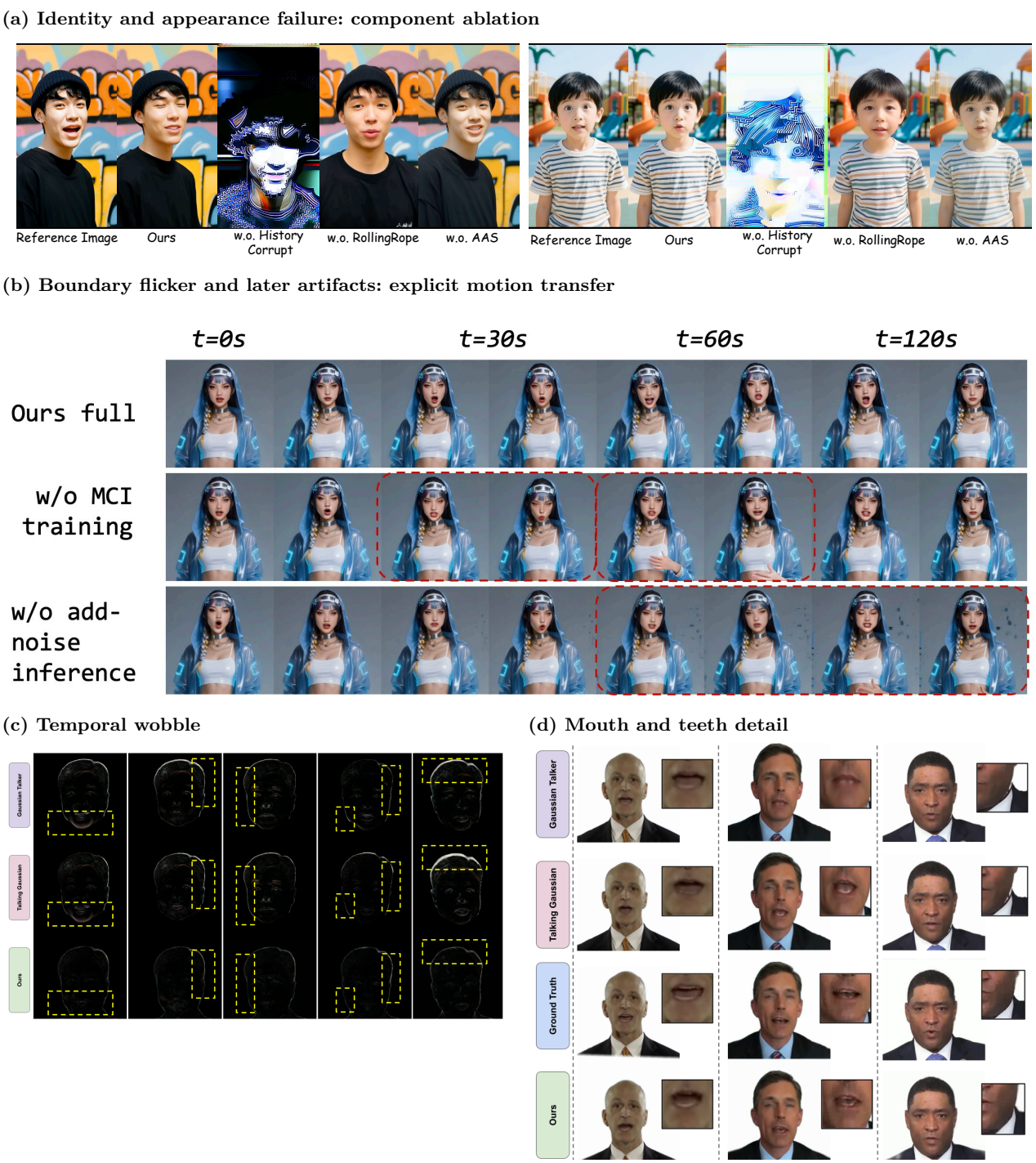



**Fig. 10** Failure examples from the original papers. (a) Live Avatar's component ablation illustrates identity and appearance degradation [81]. (b) JoyStreamer-Flash reports boundary flicker without Motion Condition Injection and later artifacts without conditioning noise; the shown times are 0, 30, 60, and 120 s [105]. (c,d) GaussianHeadTalk shows temporal wobble through image differences at ten-frame intervals, and compares mouth and teeth detail [1]. Original labels and comparison methods are retained. Static samples illustrate the reported effects but cannot show flicker directly.

### 7.5 Local Measurements Locate Costs Beyond Generation

The repeated local measurements show cost migration within concrete configurations. MuseTalk's 0.85-s generation stage sits inside a 41.53-s call, while preparation dominates DINet and LAM and output processing dominates GUAVA. FlashHead's Lite and Pro configurations likewise spend substantial time outside Gen. These identify optimization targets within each configuration; differing workloads and phase contents prevent interpreting their stage proportions as inherent route properties.

Model residency affects repeated-call cost. Forty-one configurations reload models each call, whereas 15 retain audited model objects or a worker. LiveTalk's loading still dominates its repeated short request, and no resident variant was tested for that configuration. Deployment comparisons should therefore report loading alongside generation and identify which model objects remain resident.

Repetition makes timing dispersion observable without extending the evaluated temporal horizon. Three short measured calls, with clip state rebuilt between them, do not form a continuous stability test. No matched quality or drift measurements accompany these results, so longitudinal comparisons remain those reported by the cited studies.

## 8 Datasets and Evaluation

### 8.1 Datasets for Generation and Temporal Evaluation

Datasets define the conditioning and evaluation a model can support. We compare anatomy, modalities, identity and interaction structure, media properties, splits, and continuous observation length. These properties expose task mismatches: appearance data may lack audio, reconstruction captures may lack identity variation, and dyadic clips may be too short to measure drift.

Single-person resources span distinct purposes. HDTF provides in-the-wild high-resolution audiovisual portraits, whereas VFHQ is a video-face super-resolution corpus whose source paper does not establish retained or synchronized audio [207, 246]. CelebV-HQ instead targets unconditional face-video generation and attribute editing, with manually annotated appearance, action, and emotion attributes but no source-defined audio-driving protocol [262]. MEAD adds controlled speech, synchronized views, discrete emotions, and intensity levels [192]. TalkVerse scales synchronized single-person audio–video curation and pose annotation, but its source clips remain short [202]. OpenHumanVid supplies large-scale human-centric video with caption, skeleton, and audio conditions for pretraining human video generators [106], and EchoMimicV2's half-body corpus extends conditioning to hand and gesture coverage [136]. These resources are complementary rather than interchangeable benchmarks for lip synchronization, identity, expression, or temporal behavior.

Interaction and reconstruction data introduce different supervision. ViCo pairs a visible and audible speaker with a listener's non-verbal facial response and supplies both seen-identity and identity-disjoint test regimes [260]. ResponseNet adds synchronized speaker–listener video, separated audio, and word-level transcripts for online multimodal response generation [126]. SpeakerVid-5M organizes single-speaker clips into dialogue, listening, and multi-turn structures, but its pretraining/supervised fine-tuning (SFT) tiers are quality strata rather than a documented leakage-free train/test partition [241]. Seamless Interaction records long active segments from two participants with separate audio channels and participant-level partitions, while NeRSemble supplies synchronized high-resolution multi-view head capture for sequence-specific novel-view reconstruction [2, 97]. For this survey's lenses, the relevant property of dyadic data is its *continuous-horizon* value: long uninterrupted sessions are exactly the material a stability protocol needs, independent of any interaction claim.

New interaction datasets make the preceding context part of the learning target. Face-to-Face pairs a partner's preceding observations with a subsequent audiovisual response, preserving natural reaction delays [32]. InteracVid organizes livestream material into context–query–response triplets; genuine live-chat queries and queries reconstructed from responses must remain distinguishable in evaluation [234]. These collections support context-sensitive behavior, while source-stream overlap, identity separation, and the origin of the query still determine what a test split measures.

The OmniHuman dataset and its OHBench benchmark broaden coverage to scene-level audiovisual quality, person–person and person–object relations, and individual fidelity [264]. This is distinct from the OmniHuman-1 generation method [116]. Its hierarchical evaluation helps localize failures that face-only metrics miss; benchmark clip scores still need to be complemented by continuous-session measurements when studying drift.

Table 19 distinguishes corpus scale from continuous observation length. Capture or normalized frame rates describe the data rather than generation throughput, and aggregate hours can combine many subjects, views, or short clips. The listed resources do not estab-

**Table 19** Properties of representative datasets. Media frame rate is a capture or preprocessing rate, not model throughput. Aggregate corpus duration is not a continuous generation horizon. NR denotes not reported in the primary paper.

| Dataset | Task and modalities | Scale and media | Data partition | Continuous-horizon boundary |
|---|---|---|---|---|
| HDTF [246] | In-the-wild audiovisual talking face | 362 videos; 15.8 h; 300+ subjects; 10k+ sentences; 720p/1080p | Fixed split and identity separation NR | Longest clip NR; 15.8 h is aggregate corpus time |
| VFHQ [207] | Video-face super-resolution; audio retention NR | 16,827 clips from 7,228 source videos; roughly 700–1000 px faces; $512^2$, $\times 4$ benchmark | 50 test sequences exclude corresponding training clips; identity separation NR | Tracks retain 100–2000 frames, but frame rate and total hours are NR |
| CelebV-HQ [262] | Face-video generation and 83-attribute editing; audio NR | 35,666 clips; 15,653 identities; at least $512^2$; source reports approximately 65 h in text versus 68 h in a table | Fixed identity-disjoint split NR | Each clip is 3–20 s and is truncated at 20 s |
| MEAD [192] | Controlled emotional audiovisual talking faces; seven synchronized views | 60 actors; 281,400 clips; 1920×1080 at a 30 FPS capture rate; 48-kHz audio; neutral plus seven emotions and three non-neutral intensities | Paper-specific 20% holdout; split unit and identity separation NR | 38 min 57 s per speaker aggregates many short utterances, not one session |
| ViCo [260] | Dyadic speaker audio–video to listener facial response | 483 pairs; 92 identities; 95 min 22 s; $384^2$ face crops; source video 1080p at a 30 FPS capture rate and 44.1-kHz audio | $D_{\text{test}}$ identities occur in training; $D_{\text{ood}}$ is identity-disjoint | Clips span 1–71 s; no minute-scale drift protocol |
| Response Net [126] | Dyadic video, separated audio, word-level transcripts, and facial coefficients | 696 pairs; 161 identities; 14.2 h; $1024^2$ face streams | 417/139/140 train/validation/test pairs; participant-pair disjointness stated; individual-identity disjointness NR | Clips span 27.13–863.13 s; data duration does not establish generated-session stability |
| SpeakerVid-5M [241] | Single, dialogue, listening, and multi-turn audiovisual organizations | Single branch: 5.2M clips/8.7k h; dialogue branch: 770k pairs/1.8k h; branches are not proven disjoint | Pretraining/SFT are quality tiers; VidChatBench has 500 unseen-speaker pairs, but global source/identity leakage control is NR | Basic clips span 3–14 s; stored order does not establish a continuous long-horizon sample |
| TalkVerse [202] | Single-person synchronized audio–video with 2D poses and structured annotations | 2.3M clips; 6.3k h; 720p/1080p; unified to a 25 FPS content rate | Fixed identity-disjoint split NR | Source clips span 5–50 s; aggregate hours do not establish long-session samples |
| OpenHuman Vid [106] | Human-centric video with caption, skeleton, and audio conditions | 13.2M clips reported by the source; large-scale pretraining corpus | Source-defined filtering tiers; identity-disjoint split NR | Clip-scale samples; not a continuous-session resource |
| NeRSemble [97] | Multi-view human-head capture and sequence-specific novel-view synthesis | 222 participants; 4,734 sequences; 31.7M camera images; 7.5 h; 16 views at 3208×2200 and a 73 FPS capture rate | Dataset split NR; experiments use 12 input and four held-out views within each sequence | Roughly 3 min per participant is divided among 25 short sequences; longest sequence NR |
| Seamless Interaction [2] | Dyadic speech, face, upper-body, hand, transcript, and interaction metadata | 4,065.04 h; 64,739 interactions; 5,098 sessions; 4,284 participants; dual 48-kHz audio and 2160×3840 video at a 30 FPS capture rate | Public train/dev/test plus private test; participant partition is functionally clique-level | Released active segments span 2–10 min; one-hour sessions include unreleased meta time; generation evaluation uses about 20 s |

*Abbreviations:* SFT, supervised fine-tuning; $D_{\text{test}}$ and $D_{\text{ood}}$, test and out-of-distribution subsets; 2D, two-dimensional; p, progressive-scan resolution; px, pixels; k, thousand; M, million; kHz, kilohertz; h, hours; min, minutes; s, seconds; dev, development; FPS, frames per second.

lish a matched hour-long continuous-generation benchmark. Some long-horizon evaluations therefore extend shorter inputs through repetition, as in Live Avatar [81], which limits the diversity of conditions tested. A public benchmark of continuous 10–60-minute single-subject and dyadic sessions would help measure drift under sustained, changing inputs.

Dataset governance is also part of evaluation validity. Web-derived collections report filtering and source-tracking procedures, but public availability of source videos does not by itself establish participant consent or redistribution rights [202, 207, 241, 260]. Controlled captures provide stronger acquisition records: NeRSemble reports General Data Protection Regulation (GDPR)-compliant participant agreements and deletion rights, and Seamless Interaction reports adult informed consent and compensation [2, 97]. Their primary papers nevertheless do not specify a complete dataset license. Reproducible reporting should therefore separate media provenance, consent, access terms, and technical split integrity rather than compressing them into a binary *public* field.

## 8.2 Evaluation Scope and Protocols

Evaluation separates visual quality, identity, driving fidelity, temporal fidelity, and deployment behavior (Section 2.1). Image similarity does not establish synchronization; throughput does not establish low time to first frame (TTFF) or long-session stability. Each measure must be interpreted within its tested conditions.

Four classes of evaluation measures are useful. *Full-reference measures* compare an output with a paired and aligned target. *Distributional measures* compare sets of generated and real samples. *Task-specific evaluators* estimate an attribute such as face identity or

audiovisual synchronization. *System and longitudinal measures* characterize the visible pipeline boundary and change over an evaluated session. These classes complement rather than replace one another.

### 8.3 Reference-Based and Distributional Visual Quality

Pixel errors, peak signal-to-noise ratio (PSNR), and the structural similarity index (SSIM) require a paired reference. SSIM compares local luminance, contrast, and structure before spatial aggregation [201], and its relationship to PSNR is itself protocol-dependent [72]. Its original validation concerns aligned still images; a video-level report must therefore state the frame pairing and temporal aggregation, and it should not interpret framewise SSIM as a temporal-consistency result. Learned perceptual image patch similarity (LPIPS) instead aggregates channel-normalized deep-feature differences across layers [238]; ensemble variants improve its robustness to transformations [92]. A lower value means that the candidate is closer to the paired reference under the selected backbone and calibration variant. Comparisons must retain the same network, layer set, preprocessing, crop, dynamic range, and aggregation.

The Fréchet Inception Distance (FID) fits Gaussian statistics to pretrained Inception features from real and generated image sets and computes their squared Fréchet distance [68]. Because this construction discards sample order, FID is an image-distribution measure that does not measure temporal stability. Fréchet Video Distance (FVD) applies the same statistical form to spatiotemporal features; the original study selected Kinetics-400 RGB Inflated 3D ConvNet (I3D) logits and showed that finite-sample values depend on sample count [185]. Consequently, an FID or FVD number is comparable only after the reference set, feature network and checkpoint, activation layer, preprocessing, sample count, and, for FVD, clip construction and content frame rate have been fixed. Neither distance isolates identity, lip synchronization, or deployment speed. The same protocol dependence applies to similarity scores computed with other pretrained encoders, such as CLIP-based text or image similarity [154]: the encoder checkpoint and preprocessing are part of the metric's identity.

### 8.4 Human-Specific Fidelity

Human-specific evaluators require equally explicit boundaries. ArcFace is a face-recognition representation trained with an additive angular margin [46]; it is not itself a standardized virtual-human quality score. When a paper reports ArcFace-based cosine similarity (CSIM), the report should identify the embedding model and checkpoint, face detector and alignment, reference image or set, handling of failed detections, and frame or track aggregation. Recognition accuracy reported by the ArcFace source cannot be transferred into a threshold for generated-video identity fidelity.

For audiovisual synchronization, the underlying evaluator is SyncNet, which learned a joint audio–video embedding for lip-sync assessment in the wild [37]. Wav2Lip introduced lip-sync error distance (LSE-D; lower is better) and lip-sync error confidence (LSE-C; higher is better) on top of a pretrained SyncNet [150]; the evaluator differs from the synchronization expert used to train its generator. Cross-paper comparison therefore requires the same evaluator checkpoint, audio–video pairing, temporal window, face and mouth crop, preprocessing, and averaging rule. These scores test the selected audiovisual correspondence protocol; they do not measure expression quality, identity preservation, or latency.

Landmark Distance (LMD) was defined for paired lip-motion evaluation using Dlib detections on generated and ground-truth video [26]. The original construction uses 20 lip landmarks, aligns the mean landmark positions, computes corresponding Euclidean distances, and averages over points and frames; lower is better. LMD is thus a paired landmark-accuracy measure, not a no-reference audio–lip synchronization score. Detector version, landmark subset, crop and coordinate scale, alignment, frame correspondence, failure policy, and aggregation must remain fixed for numerical comparison.

Identity, synchronization, and landmark measures cover different failure modes. They should be paired with task-appropriate motion measures only when the underlying detector, normalization, reference, and aggregation are documented. A learned evaluator's name alone is not a reproducible protocol.

Temporal alignment changes the question asked by a metric. Temporally-Aligned Evaluation uses Soft-DTW to compare feature trajectories while allowing local temporal offsets [247]. This is useful when motion has several plausible phase alignments, but an alignment that tolerates delay should not replace unwarped synchronization or measured reaction latency. The feature extractor, allowed alignment, and aggregation must remain explicit.

Fine oral appearance requires a complementary reference. RGOR uses same-person enrollment frames and oral-region supervision to preserve teeth and mouth identity [181]. Its comparison of cross-clip and target-frame references shows why paired reconstruction can

**Table 20** Properties and comparison settings of representative visual-quality measures. PSNR denotes peak signal-to-noise ratio; SSIM, structural similarity index; LPIPS, learned perceptual image patch similarity; FID, Fréchet Inception Distance; and FVD, Fréchet Video Distance. The arrows ↑ and ↓ indicate that higher and lower values, respectively, are preferable only within a matched protocol.

| Measure | Comparison unit | Reference | Minimum fields to report | Does not by itself establish |
|---|---|---|---|---|
| PSNR / SSIM ↑ | Aligned image or frame pair | Paired target | Crop and alignment, color space, dynamic range, resolution, implementation, frame aggregation | Temporal coherence, identity, lip sync, or system timing |
| LPIPS ↓ | Aligned image or patch pair | Paired target | Backbone, calibrated variant, layer set, normalization, crop, resolution, aggregation | Attribute-specific fidelity or temporal behavior |
| FID ↓ | Image sets | Real image set | Reference set, feature checkpoint/layer, preprocessing, sample count, random sampling | Sample order, temporal stability, identity, or lip sync |
| FVD ↓ | Video-clip sets | Real clip set | Video encoder/checkpoint/layer, clip length and sampling, frame rate, preprocessing, sample count | Failure localization, long-horizon stability, or runtime |

credit access to the answer. Separating enrollment from query clips and inspecting oral detail helps distinguish reference fidelity from lip synchronization or whole-face identity similarity.

### 8.5 Whole-Body, Interaction, and Cross-Scene Evaluation

Recent avatar systems expand the evaluation surface beyond face-only fidelity. Whole-body generation calls for separate face lip-sync, hand-keypoint consistency, gesture–audio alignment, body-motion diversity, face–body temporal coherence, and identity consistency across body regions. Listening-oriented systems additionally require response appropriateness, listener-motion diversity, speaking/listening transition smoothness, turn-taking responsiveness, and user preference; these should be reported separately from the latency of receiving the partner's signal. Cross-scene identity evaluation should distinguish same-scene retention from changes in background, lighting, framing, or viewpoint, because a persistent reference may preserve identity in one scene while failing under a context shift.

Temporal wobble and tracking stability form another independent axis. A render-based system can preserve a fixed identity asset while introducing jitter through tracking, deformation, Gaussian mapping, or compositing. Reports should therefore specify the temporal artifact detector, reference coordinate system, time bins, failed-detection policy, and aggregation, and should not substitute a single-frame identity score or renderer FPS for a longitudinal wobble measurement. These dimensions are complementary to ArcFace, SyncNet, and LMD and should be stratified by anatomy, interaction mode, scene constancy, and failure type [1, 36, 101].

Human-centered preference models provide another complementary view. HVEval+ collects scores and pairwise preferences for spatial quality, temporal quality, and text–video correspondence; MoE-Rater predicts these dimensions [205]. Their separation avoids treating a visually sharp frame as proof of natural motion. Such clip-level ratings characterize the presented stimuli and do not automatically quantify deterioration over a long session.

A complete protocol should disclose which of these measurements are paired, no-reference, or human-rated, together with the continuous duration and reset policy. At present, the literature reports subsets under incompatible settings rather than a unified whole-body, dyadic, cross-scene benchmark.

### 8.6 System Timing

System evaluation requires a declared boundary (Section 2.7): hardware and device count, resolution and media rate, precision, concurrency, warm-up, measurement window, and included stages. FPS must distinguish generated throughput from content or capture rate. RTF compares elapsed time with media duration; TTFF starts at a specified input event; inter-chunk latency depends on buffering; and NFE counts model calls per generated unit. FPS and RTF are convertible only with matching frame counts, rates, intervals, and boundaries. Neither determines TTFF or NFE, and mismatched workloads do not support a controlled ranking.

### 8.7 Single-H200 Runtime and Memory Measurements

We report repeated measurements of 56 configurations covering 55 methods: 11 GAN, 20 diffusion, 24 render, and one UniLS motion visualization (Tables 5, 8, 15, and 12). FlashHead Lite and Pro count as two configurations of one method. The supplied main sheet covers 50 configurations and an additional sheet contributes

**Table 21** System and temporal measurements used in virtual-human evaluation. FPS denotes frames per second; RTF, real-time factor; TTFF, time to first frame; NFE, number of function evaluations; and NR, not reported in the primary paper.

| Record | Required fields | Interpretation boundary |
|---|---|---|
| Measured frame throughput (FPS) | Frames counted, wall-clock elapsed interval, resolution, batch or concurrency, warm-up, hardware, precision, included pipeline | Establishes FPS throughput only for the disclosed generation boundary; does not establish latency or content/capture frame rate |
| RTF | Wall-clock numerator, generated-duration denominator, clip or stream length, pipeline boundary | Values below one establish faster-than-playback only for that convention |
| TTFF | Input-arrival event, buffering, first observable frame | First-frame service metric; does not establish steady-state inter-chunk latency or throughput |
| Inter-chunk latency | Successive committed-output timestamps, commitment interval, buffering, jitter summary | Steady-state service metric; does not establish TTFF or throughput |
| NFE / serial calls | Generated unit, solver steps, denoiser or vector-field calls, guidance multiplier, decoder calls | Call count does not establish wall-clock speed |
| Memory | Peak GPU memory, measurement tool, loading/warm-up policy, frame count, resolution, batch size, and device count | Use the same memory definition across methods; distinguish inference from loading and subject-specific fitting |
| Stability protocol | Evaluated continuous duration, persistent state, context bound, time bins, identity and appearance drift variables, failure counts, reset policy, and resource trajectories | A long demonstration or aggregate dataset duration does not quantify temporal stability |

six; both are included here. No native visual-AR implementation is measured. Source-paper results retain their original settings in separate tables.

*Repetition and model residency.* Each configuration runs three complete warm-ups followed by three measured calls in the same interpreter, giving 168 warm-up and 168 measured calls. Every call replays the declared input clip and produces a new output. Table times are arithmetic means of measured calls; Call also shows sample standard deviation. The accompanying files retain all six observations, medians and stage-specific sample standard deviations. Three repetitions characterize this suite, not independent processes, inputs, or long-term service variability.

Same-interpreter execution does not imply universal model residency. Forty-one configurations reload models each call while retaining process-level caches; 15 retain audited model objects or the neural GPU worker, marked *R*. The latter still repeat input preparation and rebuild clip state. StyleAvatar retains its style generator and stateless ORT sessions but reconstructs FaceVerse tracking/fitting state; Wan-Animate-2 retains the actual GPU worker as well as its controller. The first call can reuse disk caches, and three warm-ups do not establish convergence for other input shapes.

*Hardware and workload.* Each configuration uses one H200. Valid suites run across three nodes, with four CPU-affinity identifiers and numerical-library thread limits of four per configuration. Affinity identifiers need not denote independent physical cores, and host memory, storage and caches remain shared. The supplied protocol rejects attempts with foreign GPU processes or cross-device leakage; our review verifies the workbook rather than rerunning that isolation check.

Inputs, output lengths, resolution, sampling, precision, offloading and prepared assets differ. Dimensions describe the generated region or paste-back canvas, excluding comparison panels, and can exceed the internal neural crop. StyleAvatar generates two frames; LeapTalk uses one chunk and TaoMate one window. Supplied tracking, features and subject-trained models remain prerequisites where accepted by the demo, whereas LAM and StyleAvatar perform fresh tracking. Full input/sampling settings for the six additional configurations were not delivered; the appendix marks these omissions.

Actual output lengths are retained for every repetition. Talker-T2AV resets seed 42 at the generation boundary after model initialization, producing 136 frames in all six calls. Wan retains native seed $-1$ and audio/video muxing; its measured final outputs contain 35, 33 and 34 frames at $528 \times 935$. The supplied explanation attributes this variation to native shortest-stream muxing, but the corresponding server-side pre-mux evidence was not included for independent inspection. No common seed or bitwise-equivalence claim applies across methods.

*Timing boundaries.* Stage events use monotonic time and synchronization at relevant Torch CUDA or Jittor boundaries. Pre covers declared input preparation and conditioning; Gen covers the main neural path; Post covers conversion, decoding where assigned, composition and encoding. Boundaries are implementation-specific: dense-diffusion VAE decoding is generally in Post, appearance-reuse Gen usually includes controller and renderer, FLOAT retains internal encoders in Gen, and UniLS places FLAME visualization in Post. Synchronization can itself alter an asynchronous pipeline's behavior.

Load is the per-call recorded loading interval; NR means no interval is available, not zero loading cost. Call spans inference entry through synchronized return, including loading, native validation and unclassified overhead; generic output QA follows it. Stage columns use event unions within each phase. Ditto and Talk Like You also overlap phases, so their sum is not elapsed time. SyncTalk's native validation lies inside Call but outside Pre/Gen/Post. Phase unions, overlaps and validation times accompany individual calls.

Suite process time spans the entire 3+3 execution, including startup, setup, all six calls, QA and shutdown. It is neither a single-call measurement nor TTFF; dividing it by six would mix warm-up with measured execution. Main tables display Call alongside stage means. Output-commit timestamps and inter-chunk jitter were not measured.

*Memory measurement.* The main result tables report warmed memory: $M_w$ is the maximum NVML peak across the three measured calls after three complete warm-ups. The window includes each call's loading, input preparation, generation and output processing. NVML samples memory owned by the suite's GPU processes and workers approximately every 100 ms. Brief allocations may escape sampling. Torch allocated/reserved counters are reset at each call and retained separately; they include resident allocations and do not cover all Jittor, ORT or other CUDA memory.

*Metadata and interpretation.* Attention labels for 46 implementations describe separate execution observations; labels for ten configurations come from source/configuration inspection. No attention capture accompanies the timed suites themselves. Package versions and repository heads are post-run inventories, not benchmark-time captures or full instrumentation snapshots. The workbook and README provide the measurement evidence; the separate JSON, server traces, adapters and videos referenced there were not delivered. We reconciled all table means, sample standard deviations, frame sequences, phase unions and measured memory maxima against the 336 workbook records.

These measurements support stage-cost and execution-window analysis within declared configurations. Delivered output checks test file freshness, decodeability and nonconstant samples; they are not identity, synchronization or perceptual-quality measurements. Unmatched workloads and residency policies prevent a controlled algorithm ranking. Repeated short calls do not evaluate one continuous session, so matched quality, sustained service and long-session drift remain outside this experiment's evidence.

### 8.8 Longitudinal and Human Evaluation

Longitudinal evaluation measures how output quality and system behavior change during continuous generation. A reproducible protocol should specify session content and duration, initialization, retained context, and reset policy. Identity and appearance should be measured in predeclared time bins relative to an initial or short-horizon reference, alongside synchronization, motion, temporal artifacts, memory, and latency. Failures, resets, and recovery intervals require separate reporting because averages can conceal them. Ablating the active L1–L5 mechanisms under matched conditions would further distinguish their individual contributions from the behavior of the complete system.

We recommend human studies when automatic measures do not isolate visible artifacts or preference. Reports should state the question, stimuli, duration, and presentation order. They should also report rater recruitment, repeated judgments, aggregation, uncertainty, and the statistical test. A preference percentage without these fields is descriptive rather than a transferable quality scale.

### 8.9 Matching Tasks to Evaluation Data

The evaluation data should reflect the task and input conditions of each method. Audio-driven portraits need synchronized speech and face video; video-driven reenactment needs motion targets; subject-specific renderers require enrollment data; and body-generation tasks need measurements beyond the face. Table 22 summarizes these relationships. Continuous-session evaluation additionally needs a declared duration and input sequence, with repeated and varied inputs analyzed separately.

## 9 Discussion, Challenges, and Future Directions

The four routes offer different ways to distribute appearance, motion, and temporal state across a generation pipeline. Their practical value depends on the task, available resources, and evaluation conditions. The methods reviewed in Sections 3–8 motivate four common research problems: composing modules with explicit interfaces, measuring complete-system cost, maintaining consistency with bounded state, and evaluating deployment safeguards.

The synthesis in Section 7 separates measured effects from architectural deductions. The following questions build on those findings.

**Table 22** Example task–dataset pairings and suitable measurements. Continuous-session tests require sufficiently long inputs, and whole-body tasks extend evaluation beyond face-only measures.

| Task | Candidate source material | Measurements and limits |
|---|---|---|
| Audio-driven portrait | HDTF; MEAD | Identity, lip synchronization, expression/control fidelity, artifact ratings, and continuous service traces. Paired image metrics require aligned targets. |
| Video-driven reenactment | VFHQ; CelebV-HQ | Pose/landmark target error, identity, temporal artifacts, and disocclusion. Unpaired cross-identity reenactment lacks a pixel-matched RGB target. |
| Subject-specific rendering | NeRSemble; declared subject capture | Novel-view/reenactment quality, tracking/deformation wobble, capture and fitting cost. Multi-view capture is an enrollment condition. |
| Selected upper/full-body animation | OpenHumanVid; task-specific video/pose sequences | Face synchronization, body/hand keypoint fidelity where annotations permit, gesture–audio correspondence, and face–body consistency. Face identity scores do not cover hands or body motion. |

### 9.1 Composition Requires Explicit Interfaces

Hybrid systems divide work among condition or motion models, a visual owner, a decoder or renderer, and a compositor. The ownership rule in Section 2.3 assigns the route [22, 64], while all modules remain within the system boundary. Improving one module can change control or synthesis cost while leaving encoding, decoding, and transport on the critical path.

Composition requires both a representation interface and a state-update policy. Each module should specify its coordinate system, identity reference, subject-specific assets, retained history, reset behavior, and recovery procedure. These fields determine when output becomes available and which errors can propagate. For example, Teller's audio buffering precedes motion prediction and rendering, while LivePortrait's reused appearance still requires motion estimation and decoding [62, 254].

Compact motion interfaces can reduce the state presented to a dense visual owner, but they also restrict appearance changes to those expressible by the interface. Dense latent interfaces offer more freedom while making identity ownership and failure localization harder to analyze. These are testable architectural trade-offs rather than universal advantages of a route. A falsifiable research question is whether a hybrid can expose typed representation, state, and recovery contracts such that the latency and failure contribution of every module is reproducible in isolation and in composition.

### 9.2 Complete-System Cost and Timing

Published FPS, latency, real-time factors, NFE, and speedup ratios use different boundaries; the route tables retain their reported settings. Renderer-only timing does not establish system latency [62, 81]. Our repeated measurements of 56 H200 configurations add stage timings, sample dispersion and warmed memory peaks. DINet and LAM expose preparation costs, while GUAVA exposes export costs. Model residency, matched inputs and quality targets remain necessary considerations for a controlled method comparison.

Deployment records should separate enrollment from runtime and time every stage through I/O. User-visible measures include TTFF, inter-commit latency and jitter, throughput, memory, and recovery cost. Section 7 shows how decoder choice changes quality and throughput, while pipelining can increase throughput and first-output delay together. Component profiles should test where the bottleneck moves with resolution, chunk size, concurrency, and session duration, including preparation and recovery.

### 9.3 Bounded-State Long-Horizon Generation

The five patterns in Section 2.8 provide a common vocabulary for mechanisms whose evaluation remains uneven. The reviewed diffusion systems contain examples of all five, including Live Avatar's sampled observations beyond an hour under looped audio [81]. Render-based systems reuse an identity asset but can drift through their controllers or rendering pipeline; DyStream's 180-s pose-drift ablation provides a limited longitudinal example [22]. Native visual AR explicitly retains history, yet the reviewed systems lack a complete joint stability protocol. Framewise GAN generators avoid recursive visual feedback when earlier outputs are not reused, but can still exhibit jitter or inherit controller errors.

This comparison suggests studying which mechanisms transfer across representations. Training with generated histories could be evaluated in both native visual AR and recurrent motion controllers. The anchor-regulated updates of TaoMate and the selection policy of InteractiveAvatar offer concrete alternatives to fixed windows [56, 173]. Comparisons with compression

and resets should now test recovery after corrupted history and requested reference changes. DynaForcing and Routed Forcing also motivate testing motion amplitude and diversity alongside appearance consistency [82,175]. A matched study should isolate each mechanism's effect on identity, synchronization, motion diversity, and legitimate scene changes over predeclared time bins. The reviewed studies cover different tasks and durations, leaving sustained operation under varied conditions an open challenge.

*Audio-to-deformation and expanded Stable axes.* Recent audio-driven Gaussian systems expose a shift from the conventional audio → pose or SMPL → renderer cascade toward direct audio → Gaussian or particle deformation → renderer paths. This can preserve high-frequency facial, hand, and body dynamics that an intermediate pose interface may quantize, but it moves complexity into learned deformation fields and their tracking boundary. The resulting Stable problem is multidimensional: appearance identity, motion and deformation, audio–visual synchronization, interaction response, and system/resource behavior can fail independently. Generalizable Gaussian priors reduce enrollment cost but introduce adaptation and out-of-distribution trade-offs; fixed Gaussian scenes anchor appearance but do not prevent wobble, tracking failure, or compositing drift. These systems motivate joint evaluation of enrollment and runtime costs with time-binned identity, motion, synchronization, wobble, failure, and recovery measurements [1,4,74,101,115].

## 9.4 Route Selection Under Deployment Constraints

Table 23 summarizes overlapping suitability conditions. The design guidance of Sections 3–6 supports a first set of conditional guidelines. These summarize documented design options under stated constraints; they do not rank individual methods, and each guideline inherits the measurement caveats of its underlying sources.

*When per-subject enrollment is acceptable*, render-based systems offer reusable appearance and reported high-throughput or wearable configurations [25, 206]. Selection should account for capture and fitting cost, followed by tracking, deformation, and compositing over the intended session duration.

*When one-shot operation is required*, compact motion generation over reused appearance provides a practical design pattern [62, 67, 210]. The motion module and renderer must both fit the latency budget, while deformation and disocclusion determine which movements the appearance model can reproduce.

*When changes extend beyond a reusable appearance model*, dense diffusion offers mechanisms for synthesizing broader visual content. The reviewed streaming examples combine distillation, memory, and substantial serving resources [81, 165]. Persistent anchors can also restrict scene changes, so the intended variability must be evaluated directly. Component profiling should determine whether further optimization targets the sampler, decoder, or conditioning path.

*When the edit is confined to a small region*, feed-forward GAN synthesis can reduce the amount of generated content [150, 242]. Evaluation should include compositing seams, synchronization, and framewise jitter as well as runtime.

*When explicit visual-history modeling is central*, native visual AR offers a direct formulation. Its deployment evaluation remains limited in the reviewed studies: Table 11 does not contain a system with both full-pipeline timing and a joint time-binned stability evaluation. The representation, local emission head, and state policy should therefore be validated for the intended application.

## 9.5 Trustworthy Deployment

Fast and Stable systems sharpen deployment risks: low-latency synthesis enables live impersonation, and long-horizon stability with cached identity state extends it. Trustworthiness is not an additional scalar metric. The NIST AI 100-4 report separates provenance-data tracking from synthetic-content detection and discusses watermarks, metadata, filtering, and red-teaming as controls with different lifecycle positions and failure surfaces [20]; it provides no avatar-specific effectiveness or runtime measurement. The C2PA specification defines signed claims, content bindings, and validation states, including segment-level paths for live video [40]. These mechanisms bind a signing credential to covered media bytes and expose specified integrity failures; they do not establish factual truth, verify the depicted identity, or prove authorization to use a likeness, and they carry manifest-stripping, key-management, and privacy risks [38, 39].

Empirical safeguards have equally important boundaries. DeepfakeBench exposes generalization gaps across manipulation types in frame-level evaluation [216], and Video Seal demonstrates faster-than-playback video watermarking under a stated offline boundary without adaptive-attack coverage [55]. A watermark hit, a detector score, and a valid provenance record are different observations; none alone proves authorization or harmlessness. A deployment evaluation should declare the attacker model, benign platform

**Table 23** Requirements that guide method selection. Suitability depends on the intended task, available resources, and supported controls.

| Requirement | Candidate mechanism | Additional condition to check |
|---|---|---|
| Regional lip editing | Wav2Lip+GAN, MuseTalk | Source pixels remain usable; inspect seams, synchronization, and jitter. |
| Reusable identity and controllable pose | Fields, Gaussians, feature warps | Enrollment is feasible *and* the representation supports the intended appearance, views, and motion. |
| One-shot portrait animation | LivePortrait, VASA-1, LAM | Test disocclusion, pose limits, and complete controller–renderer timing. |
| Appearance or scene changes | Dense visual generation | Validate the specific change. A static-scene system such as Live Avatar does not establish open-ended scene editing. |
| Explicit visual-history control | Native visual AR | Profile local sampling and decoding; test long-rollout errors and resets. |

transformations, operating point, authorization and revocation procedures, and the latency cost of each control on the serving path. The last point connects trust directly to this survey's Fast accounting, since provenance signing and watermarking must fit inside the same per-chunk budget the route mechanisms optimize.

### 9.6 Conclusion

Fast and Stable virtual-human generation depends on the representation and computation of appearance, motion, and temporal state. GANs emphasize feed-forward regional synthesis; diffusion reduces sampling work and reuses computation; visual AR models evolving appearance; and rendering reuses identity-specific assets. Decoder choice, serving schedules, and state policies connect efficiency with temporal quality. Our single-H200 study of 56 configurations and 55 methods uses three measured calls after three complete warm-ups. Preparation, decoding, export, and loading can outweigh generation, making stage boundaries and residency essential to cost interpretation. The different workloads and absence of matched quality or continuous-session measurements leave the joint Fast–Stable comparison open; sustained, controlled evaluation is the next step.

## Data Availability Statement

The accompanying measurement files contain 56 single-H200 configurations, all 336 calls, full-precision summaries and sample standard deviations, input/stage descriptions, and software/attention provenance (Section 8.7). The supplied workbook and README are the measurement sources. Referenced server traces, adapters, videos, and separate JSON were unavailable for independent inspection; matched quality and longitudinal evaluations are not included. The public companion provides the measurements and a searchable literature catalogue.

## A Implementation Settings for the Local Measurements

Tables 24, 25, and 26 summarize inputs, settings, stage boundaries and model residency for all 56 configurations. Italic text identifies input settings; upright text describes stage boundaries and prerequisites. *R* denotes retained models or workers; other configurations reload each call. The accompanying CSV/JSON files retain all 336 calls, full-precision means and sample standard deviations, phase unions and overlaps, sampled memory peaks, separate PyTorch counters, and suite process times. Timing definitions follow Section 8.7. Full sampling/decoder settings and operator-level traces for the six additional diffusion configurations were not included in the delivery; these omissions remain explicit.

Moore-AnimateAnyone and face-vid2vid-unofficial are third-party implementations; RAD-NeRF is described by its repository as a PyTorch reimplementation. Repository heads and package versions are supplementary post-run observations, while attention labels distinguish historical execution evidence from current source/configuration inspection. They do not establish the kernel dispatch of the measured calls or a complete instrumentation snapshot. Supplied tracking and trained assets remain distinct from timed preparation.

**Table 24** Input settings and stage boundaries for the 11 GAN configurations.

| Configuration | Input / configuration and stage boundary |
|---|---|
| Wav2Lip+GAN$^R$ | *EDTalk bundled input; Wav2Lip CLI.* Face/audio preparation; crop synthesis; composition and encoding. |
| MuseTalk$^R$ | *First supplied test.yaml case; audited models retained.* Face coordinates, audio/image features and output processing repeated for every clip. |
| TalkLip$^R$ | *Supplied inf_demo input.* Explicit media, neural generation, and output stages. |
| Live Speech Portraits$^R$ | *Default May example.* Supplied pretrained metadata; media preparation and output writing included. |
| EDTalk$^R$ | *EDTalk-A example.* Media preparation, neural prediction, and output processing separated. |
| VideoReTalking$^R$ | *README example; audited models retained.* Fresh landmarks, coefficients and stabilization; restoration/encoding in Post. |
| DINet$^R$ | *Bundled test4; supplied landmarks.* Offline OpenFace landmark extraction excluded. |
| MakeItTalk | *One supplied Obama/audio pair.* Audio-to-landmarks and image translation in Gen; tensor preparation in Pre. |
| PC-AVS$^R$ | *demo_vox options and demo.csv.* Media Pre; neural Gen; restoration and encoding Post. |
| EAT$^R$ | *One supplied reference.* Cropped image, image latent, and driving features supplied; offline preparation excluded. |
| HyperReenact$^R$ | *README source/target pair.* Source preparation, neural generation, and output processing separated. |

**Table 25** Input settings and stage boundaries for the 20 diffusion configurations.

| Configuration | Input / configuration and stage boundary |
|---|---|
| StableAvatar | *50 steps; full model loading.* Audio/image/text encoders in Pre; denoising Gen; VAE decoding Post. |
| InfiniteTalk | *81-frame, one-clip mode; 40 steps.* Default offloading retained; demand-driven device transfers in Gen; VAE decoding Post. |
| LiveTalk | *Default inference script/config.* Encoders Pre; KV initialization and denoising Gen; VAE decoding Post. |
| Wan-Animate-2$^R$ | *Distilled demo, 10 steps; persistent GPU worker.* Worker timing; final frames 35/33/34, cropped to 528 by 935; seed -1 retained. |
| AvatarForcing-SW | *I2V; length setting 33, 129 output frames.* Sliding-window denoising and cache operations Gen; VAE decoding Post. |
| EverAnimate | *One clip; supplied pose/face videos.* Offline pose/face extraction excluded; conditioning Pre; composite-video export Post. |
| LeapTalk | *max_chunks=1; original compile/offload.* History re-encoding Pre; bridge sampling Gen; TAE/VAE Post; version-check compatibility override. |
| Avatar-Forever | *33 output frames; latent upsampler enabled.* Lazy loads in Load; AR diffusion and upsampler Gen; streamed VAE Post. |
| TaoMate | *First prompt segment; one window.* Text conditioning Pre; joint AV Gen; audio/video VAE decoding Post. |
| Moore-AnimateAnyone | *Third-party; first case, 16 frames.* Reference processing and denoising Gen; VAE decoding and comparison-video save Post. |
| SoulX-LiveAct | *245 frames; 3 steps; T5 on CPU; audio CFG 1.* Models reload; SDPA fallback; compile, FP8 KV and FP8 GEMM disabled. |
| SoulX-FlashTalk$^R$ | *250 frames; 4 steps; retained model/VAE/audio encoder.* Compile/offload disabled; audio and reference/motion preparation repeated. |
| SoulX-FlashHead Lite$^R$ | *250 frames; 4 steps; retained model/VAE/audio encoder.* Compile/offload disabled; Lite configuration, with new input state each call. |
| SoulX-FlashHead Pro$^R$ | *250 frames; 4 steps; retained model/VAE/audio encoder.* Compile/offload disabled; Pro configuration, with new input state each call. |
| Follow-Your-Emoji-Faster | *16 frames, 512 by 512, 7 FPS playback.* Main generated panel only; full sampling settings and stage trace not supplied. |
| Hallo-Live | *21 frames, 992 by 512, 24 FPS playback.* Model reload each call; full sampling settings and stage trace not supplied. |
| LightningCP | *16 frames, 512 by 512; prepared mask input.* Model reload each call; full sampling settings and stage trace not supplied. |
| Lip Forcing | *9 frames, 512 by 512; fresh face cache per call.* Model reload each call; full sampling settings and stage trace not supplied. |
| Live Avatar | *45 frames, 704 by 384; one H200.* Model reload each call; distinct from source multi-GPU serving configurations. |
| PersonaLive | *16 frames, 512 by 512; model reload each call.* Full sampling/decoder settings and stage trace not supplied. |

**Table 26** Settings for the 24 appearance-reuse configurations and the motion visualization.

| Configuration | Input / configuration and stage boundary |
|---|---|
| StyleAvatar$^R$ | *Two Lizhen frames; style generator and ORT retained.* Fresh FaceVerse state/fitting; Torch allocator cap 70%; Jittor allocations separate. |
| AD-NeRF | *Released head/torso assets; 16 poses.* No training; supplied audio/poses; head and torso rendering Gen. |
| RAD-NeRF | *PyTorch reimplementation; Obama demo.* Supplied pose/audio features; ray preparation Pre; test_step Gen. |
| ER-NeRF | *Released checkpoint; 16 poses.* Minimal inference CLI adapter; supplied pose/audio features; no GT evaluation. |
| SyncTalk | *May validation; 16 frames; supplied features.* Native validation in Call and memory, outside the three inference phases. |
| GPAvatar | *Tracked-driver video demo; 16 frames.* Tracking supplied; neural rendering Gen; comparison-panel composition Post. |
| LAM | *Fresh FLAME tracking directory.* Tracking and auxiliary initialization Pre; infer_single_view Gen; PLY/media output Post. |
| FOMM | *TPS bundled input; original FOMM weights.* Decode/resize Pre; motion and generator Gen; output encoding Post. |
| TPS Motion Model | *Bundled demo assets.* Decode/resize Pre; motion and generator Gen; output encoding Post. |
| LivePortrait | *Default bundled inputs.* Landmarks Pre; motion/warp/retargeting Gen; paste-back and encoding Post. |
| LIA | *Original Vox demo; 10-fps video.* Input preparation, generator inference, and output encoding separated. |
| FLOAT | *README portrait/audio example.* Internal audio and motion encoders remain in Gen. |
| Ditto | *README PyTorch implementation.* Thread operators exclude queue waits; smoothing/paste-back Post; phase overlap retained. |
| MoDA | *README defaults.* Audio/portrait conditioning Pre; motion diffusion and rendering Gen; smoothing/paste-back Post. |
| JoyVASA | *Human demo.* Landmarks Pre; motion/warp/rendering Gen; paste-back and encoding Post. |
| DyStream | *Sample 1 dyadic conversation callback.* Motion prediction and visualization decoder in Gen; composite/audio export Post. |
| Talker-T2AV | *Bundled identity, voice and text; generation seed 42.* All six outputs 136 frames; speech/motion AR and LIA rendering Gen; waveform decode Post. |
| Talk Like You | *Preset192; Stage0 neutralization retained.* Threaded motion/render operators Gen; queue waits excluded; phase overlap retained. |
| SplattingAvatar | *Released female3 asset; 16 refined poses.* No training; SMPL/deformation/splatting Gen; JPEG and wrapper MP4 export Post. |
| 3DGS-Avatar | *Released ZJU377 checkpoint; 16 dance poses.* No training; view preparation Pre; rendering Gen; wrapper packaging at 25 fps Post. |
| GAGAvatar | *README video driver.* Driver/portrait tracks supplied; forward_expression Gen; comparison output Post. |
| GUAVA | *Tracked self-reenactment example.* Tracking supplied; internal reconstruction/render warm-ups kept in Gen; PLY/media export Post. |
| UniLS | *Short audio; FLAME visualization.* Motion prediction Gen; mesh decoding, rendering, smoothing, and audio/video export Post. |
| SDTalk | *Identity133; source repository SDGTalk.* Portrait/driver tracks and cached audio supplied; neural forward Gen; composition/mux Post. |
| face-vid2vid-unofficial | *Unofficial SPADE implementation.* Bundled driving pair; motion/generator Gen; output transfer and encoding Post. |

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